Unit – IV
ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES

Pollution: Any atmospheric condition, in which certain substances are present in such concentration that they can produce undesirable effects on man and his environment.
Or
Pollution is defined as the excess discharge of any substance into the environment which affects adversely quality of environment and causing damage to human, plants and animals.
Pollutant: Any substance which causes pollution called as pollutant.

4.1 Classification of Pollutants:

The classification of pollutants is done from different points of view.
I. Depending upon their existence in nature pollutants are of two types, namely quantitative and qualitative pollutants.
a) Quantitative pollutants: These are those substances normally occurring in the environment, who acquire the status of a pollutant when their concentration gets increased due to the unmindful activities of man. For example, CO_2, if present in the atmosphere in concentration greater than normal due to automobiles and industries, causes measurable effects on humans, animals, plants or property, then it is classifies as a quantitative pollutant.
b) Qualitative pollutant: These are those substances which do not normally occur in nature but are added by man, for example, insecticides.

II. Depending upon the form in which they persist after being released into the environment, the pollutants are categorized into two types, namely primary and secondary pollutants.

a) Primary pollutants: These are those which are emitted directly from the source and persist in the form in which they were added to the environment. Typical examples of pollutants included under this category are ash, smoke, fumes, dust, nitric oxide, sulphur dioxide, hydrocarbons etc.
b) Secondary pollutants: These are formed in the atmosphere by chemical interactions among primary pollutants and normal atmospheric constituents. Pollutants such as SO_3, NO_2, PAN, O_3, aldehydes, ketones, and various sulphate and nitrate salts include in this category.

III. From the ecosystem point of view, i.e., according to their natural disposal, pollutants are of two types:

a) Bio-degradable pollutants: These are the pollutants that are quickly degraded by natural means. Heat or thermal pollution, and domestic sewage are considered in this category as these can be rapidly decomposed by natural processes or by engineered systems such as municipal treatment, plants etc.

b) Non-degradable pollutants: These are the substances that either do not degrade or degrade very slowly in the natural environment. These include mercury salts, long chain phenolic chemicals, DDT and Aluminium cans etc.

Classification of pollution:

On the basis of type of environment being polluted, pollution classified into- Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, Nuclear pollution etc.

4.2. AIR POLLUTION

Air pollutants are two types-

a. Primary pollutants: are those that are emitted directly from source. Sources are steel mills, power mills, oil refiners, paper and pulp industries, automobiles etc.

Ex. Particulate matter such as ash, smoke, dust, fumes, mist and spray; inorganic gases as SO_2, H_2S, NO, NH_3, CO, CO_2, HF, Olifinic and aeromatic hydrocarbons and radio active compounds.

b.` Secondary pollutants: These are formed in the atmosphere by chemical interactions among primary pollutants and normal atmospheric constituents. Pollutants such as SO3, NO_2, PAN, O_3, aldehydes, ketones, and various sulphate and nitrate salts include in this category.

4.2.1.Sources of Air pollution:

Air pollution results from gaseous emissions from mainly industry, thermal power stations, automobiles, domestic combustion etc.

1. **Industrial chimney wastes:** There are a number of industries which are source of air pollution. Petroleum refineries are the major source of gaseous pollutants. The chief gases are SO2 and NOx. Madhura- based petroleum refinery is posing threat to Taj Mahal in Agra and other monuments at Fatehpur Sikkri complex. Cement factories emit plenty of dust, which is potential health hazard. Stone crushers and hot mix plants also create a menace. The SPM levels in such areas of stone crushing are more than five times the industrial safety limits. There are also chemical manufacturing industries which emit acid vapors in air. During last 10 years the number of industrial units in Delhi, increased from about 20,000 to 55,000. Nearly 40,000 of these are located in predominantly residential areas.

2. **Thermal Power stations:** The coal consumption of thermal power plants is several million tones. The chief pollutants are fly ash, SO2 and other gases and hydrocarbons.

   The three thermal power stations at the Indraprastha Estate, Rajghat and Badarpur in Delhi are the main source of air pollution. The Indraprastha plant daily consumes

3,500 to 4,000 tons of coal when all the five units function. Badarpur, the largest consumes daily about 10,000 tones of coal.

3. **Automobiles:** The ever increasing vehicular traffic density posed continued threat to the ambient air quality. There are over 300 million cars, trucks and buses the world over and their number increasing rapidly. In all the major cities of the country (India) about 800 to 1000 tones of pollutants are being emitted into the air daily, of which 50% automobile exhausts. In the major metropolitan cities, vehicular exhaust accounts for 70% of all CO, 50% of all hydrocarbons, 30-40% of all oxides and 30% of all SPM.

The various points of emission from automobiles are 1. Exhaust system 2. Fuel tank and 3. Crankcase. The exhaust produces many air pollutants including unburnt hydrocarbons, CO, NOx and lead oxides. There are also traces of aldehydes, esters, ethers, peroxides and ketones which are chemically active and combine to form smog in presence of light.

4.2.2. Major air pollutants sources and effects:

1. Particulate matter: Particulate refers to all atmospheric substances that are not gases. They can be suspended droplets or solid particles or mixtures of the two. This size ranging from 100micro meters don to 0.1 micrometer and less.

Particulates classifies as-

Dust: Size 1 to 200 micrometers. These are formed by natural disintegration of rock and soil or by the mechanical process of grinding and spraying. They have large setting velocities and are removed from the air by gravity and other inertial process.

Smoke: Size 0.001 to 1 micrometer. This can be liquid or solid and are form by combustion or other chemical process.

Fumes: Size 0.1 to 1 micrometer. There are solid particles normally released from chemical or metallurgical processes in industries.

Mist: Size < 10 micrometer. It is made up of liquid droplets. These are formed by condensation in the atmospheric or released from industrial operations.

Effects: Respiratory problems- Bronchitis, asthma, irritation etc.

2. Oxides of sulfur:

SO_2 is a colorless gas with a characteristic, sharp, pungent odor. It is moderately soluble in water forming weak acid H_2SO_3. It is oxidized slowly in clean air to SO_3 In polluted atmosphere SO_2 reacts photochemically or catalytically with other pollutants or normal atmospheric constituents to form SO_3, H_2SO_4 and salts of H_2SO_4.

Effects: SO_2 and moisture can accelerate the corrosion of steel, copper, zinc and other metals. H_2SO_4 mist in the atmosphere causes deterioration of structural monuments or materials such as marble and lime stone. Paper also discolored by SO_2.

8 – 12 ppm of SO_2 - immediate throat infection,10 ppm SO_2 – eye irritation,
20 ppm – immediate coughing.

Sources: Chemical industries, metal smelting, pulp and paper mills, oil refineries.

3. Nitrogen Dioxide: Sources are natural and man made. Natural sources of nitrogen dioxides are high energy radiation, biological and non biological activities etc. Man made nitrogen dioxide coming from incineration process, pesticide industries, automobiles etc.

Effects: NO_2 combine with hydrocarbons to form photochemical smog its cause most damage to human health. In lungs NO_2 converted to nitrous and nitric acids which are highly irritating and cause damage to the lung tissues. It causes acid rain and cancer.

4. Carbon monoxide: Incomplete combustion of carbon fuels in automobiles, industries etc. The main sources of CO in the urban air are smoke and exhaust fumes of many devices burning of coal, gas or oil.

Effects: CO react with blood hemoglobin to form carboxihemoglobin.

1-5% COHb in blood –reduction in oxygen carrying capacity of blood, 30-40% COHb in blood- Severe head ache, vomiting, 50-60% COHb in blood- Coma,

70-80% COHb in blood – Fatal coma (death).

5. Hydrocarbons: Sources are automobiles, spray paintings, inks etc.

Effects: Skin, nose, throat, eye irritation. Cause cancer in human beings.

4.2.3. Ambient air quality standards:

The National Ambient Air Quality Monitoring (NAAQM) network is operated through the respective States Pollution Control Boards, the National Environmental Engineering Research Institute (NEERI), Nagpur and also through the CPCB.

The pollutants monitored are Sulphur dioxide (SO_2), Nitrogen dioxide (NO_2) and Suspended Particulate Matter (SPM) besides the meteorological parameters, like wind speed & direction, temperature and humidity. In addition to the three conventional parameters, NEERI monitors special parameters, like Ammonia (NH_3), Hydrogen Sulphide (H_2S), Respirable Suspended Particulate Matter (RSPM) and Polyaromatic Hydrocarbons(PAH).

Based on Annual Mean Concentration (microgram per cubic meter of ambient air) of SO_2, NO_2 and SPM and the Notified Ambient Air Quality Standards, the Ambient Air Quality Status is described in terms of Low (L), Moderate (M), High (H) and Critical (C) for Industrial (I), Residential and mixed use (R) areas of Cities/Towns in different States/UTs.

National Ambient Air quality Standards (NAAQS)

POLLUTANTS	Time Weighted Average	Concentration of Ambient Air			
		Industrial Area in µg/m3	Residential Rural and other area in µg/m3	Sensitive area in µ/m3	Method of Measurement
Sulphur Dioxide (SO2)	Annual Average	80	60	15	Improved west and Gacke Method
	24 hours	120	80	30	Ultraviolet fluorescence

Oxides of Nitrogen (NO2)	Annual Average 24 hours	80 120	60 80	15 30	Jacab Hochheister modified (Na-Arsentire method Gas Phase Chemilumine Scene
Suspended Particulate Matter (SPM)	Annual Average 24 hours	360 500	140 200	70 100	High Volume sampling (average flow rate not less than 1.1 m^3/minute)
Respirable Particulate Matter (size Less than 10µm) RPM	Annual Average 24 hours	120 150	60 100	50 75	Respirable particulate matter sampler
Lead as Pb	Annual Average 24 hours	1.0 1.5	0.75 1.0	0.50 0.75	AAS method after sampling using EPM 2000 or equivalent filter paper
Carbon Monoxide	8 hours 1 hour	5.0mg/m3 10.0mg/m3	2.0mg/m3 4.0mg/m3	1.0mg/m3 2.0mg/m3	Non dispersive infrared spectroscopy

Annual Average : Annual Arithmetic Mean of minimum 104 measurements in a year taken twice a week 24-hourly at uniform interval
24 Hours Average : 24-hourly/8-hourly values should be met 98% of the time in a year. However 2% of the time, it may exceeded but not two consecutive days.

1. The levels of air quality necessary with an adequate margin of safety, to protect the public health, vegetation and property.
2. Whenever and wherever two consecutives values exceeds the limit specified above for the respective category, it shall be considered adequate, reason to institute regular/continuous monitoring and further investigations.

4.2.4. Control of air pollution:

The following methods are most effective for dealing with the control of air pollution.

a. Source correction methods
b. Pollution control equipment
c. Diffusion of pollutants in air
d. Vegetation
e. Zoning

A) Source correction methods:

Industries make a major contribution towards causing air pollution. Formation of pollutants can be prevented and their emission can be minimized at the source it self.
The source correction methods are:

I) Substitution of raw materials: If the use of a particular raw material results in air pollution, then it should be substituted by another purer grade raw material which reduces the formation of pollutants. Thus,
i)Low sulphur fuel which has less pollution potential can be used as an alternative to high sulphur fuels, and
ii) Comparatively more refined liquid petroleum gas (LPG) or liquefied natural gas (LNG) can be used instead of traditional high contaminant fuels such as coal.

II) Process modification: The existing process may be changed by using modified techniques to control emission at source. For example-
i) If coal is washed before pulverization, then fly-ash emissions are considerably reduced.
ii) If air intake of boiler furnace is adjusted, then excess fly-ash emission at power plants can be reduced.

III) Modification of existing equipment: Air pollution can be considerably minimized by making suitable modifications in the existing equipment.
i) For example, smoke, CO and fumed can be reduced is open hearth furnaces are replaced with controlled basic oxygen furnaces or electric furnaces.
ii) In petroleum refineries, loss of hydrocarbon vapors from storage tanks due to evaporation, temperature changes or displacement during filling etc. can be reduced by designing the storage tanks with floating roof covers.

IV) Maintenance of equipment: An appreciable amount of pollution is caused due to poor maintenance of the equipment which includes the leakage around ducts, pipes, valves and pumps etc. Emission of pollutants due to negligence can be minimized by a routine check up of the seals and gaskets.

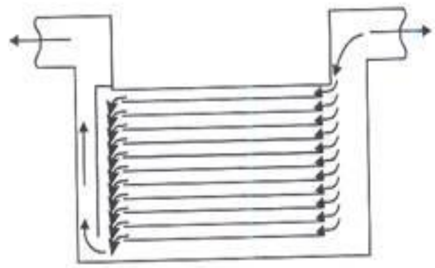
B) Pollution Control equipments:
Sometimes pollution control at source is not possible by preventing the emission of pollutants. Then it becomes necessary to install pollution control equipment to remove the gaseous pollutants from the main gas stream.
The pollutants are present in high concentration at the source and as their distance from the source increases they become diluted by diffusing with environmental sir.
Pollution control equipments are generally classified into two types:

I) Control devices for particulate contaminants
II) Control devices for gaseous contaminants.

I) Control devices for particulate contaminants:

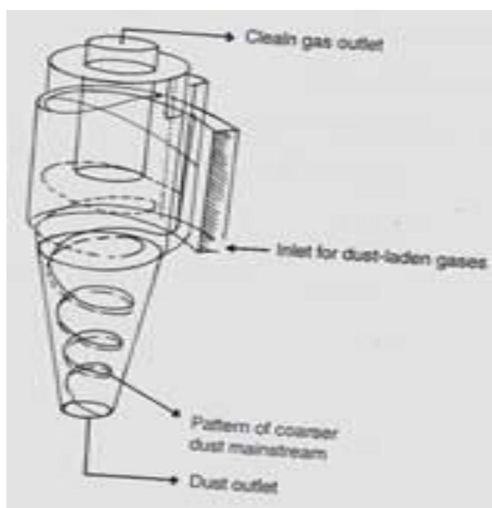
i) Gravitational settling chambers: In gravitational settling chambers, the particulate matter is settled by its own weight by lowering the flue gas velocity. The settling chamber may consist of a simple balloon duct, an expansion chamber with dust hopper or dust settling chamber.
The multiple tray settling chamber as shown in figure below represents one of the first attempts to increase collection efficiency by reducing the vertical settling distance by

using multiple shelving. Because the settling rate of dust decreases with increasing turbulence of the gas, the velocity of the gas stream is usually kept as low as possible.



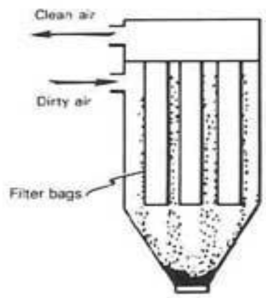
Settling chambers are usually installed as pre-cleaners to remove large and agglomerated particles, which can clog small diameter cyclones and other dust cleaning equipments. It essentially consists of a chamber in which the velocity of the carrier gas is decreased so that particles in the gas settle down by gravity.

ii) Cyclone separators (reverse flow cyclone): A cyclone as shown in figure below is an inertial separator without moving parts and separates particulate matter from carrier gas by transforming the velocity of inlet stream into a double vortex confined within the cyclone.



In the double vortex, the entering gas spirals downward at outside and spirals upward at the inside of cyclone outlet. The particulates, because of their inertia, move towards the outside wall from which they are led to receiver i.e. conical portion.

iii) Fabric filters (Bag house filters): Commonly known as bag house, fabric collectors use filtration to separate dust particulates from dusty gases. They are one of the most efficient and cost effective types of dust collectors available and can achieve a collection efficiency of more than 99% for very fine particulates. Dust-laden gases enter the bag house and pass through fabric bags that act as filters. The bags can be of woven or felted cotton, synthetic, or glass-fiber material in either a tube or envelope shape. The high efficiency of these collectors is due to the dust cake formed on the surfaces of the bags.



Baghouse

The fabric primarily provides a surface on which dust particulates collect through the following four mechanisms:

-Inertial collection - Dust particles strike the fibers placed perpendicular to the gas-flow direction instead of changing direction with the gas stream. -Interception - Particles that do not cross the fluid streamlines come in contact with fibers because of the fiber size. -Brownian movement – Sub micro meter particles are diffused, increasing the probability of contact between the particles and collecting surfaces. -Electrostatic forces - The presence of an electrostatic charge on the particles and the filter can increase dust capture.

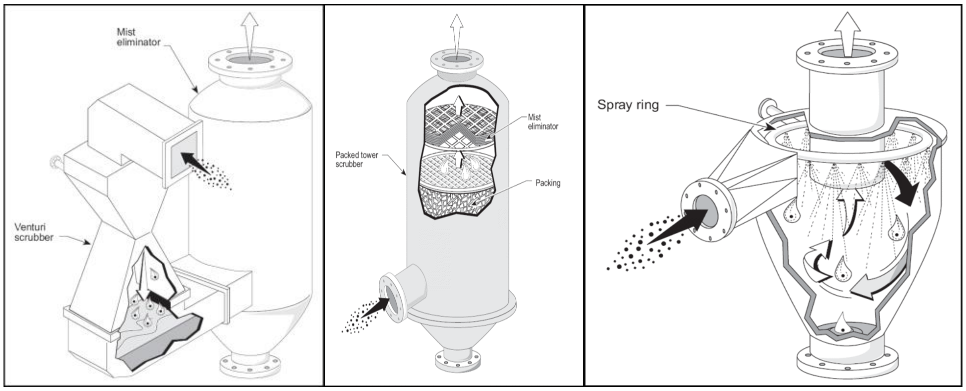
Combination of these mechanisms result in formation of dust cake on the filter, which eventually increases the resistance to gas flow. The filter must be cleaned periodically.

iv) Wet collectors (scrubbers): In wet collectors or scrubbers the particulate contaminants are removed from the polluted gas stream by incorporating the particulates into liquid droplets. Common wet scrubbers are: Spray towers, Venturi scrubbers, Cyclone scrubber

In a wet scrubber, the polluted gas stream is brought into contact with the scrubbing liquid, by spraying it with the liquid, by forcing it through a pool of liquid, or by some other contact method, so as to remove the pollutants.

The design of wet scrubbers or any air pollution control device depends on the industrial process conditions and the nature of the air pollutants involved. Wet scrubbers remove dust particles by *capturing* them in liquid droplets. Wet scrubbers remove pollutant gases by *dissolving* or *absorbing* them into the liquid.

Generally, the inlet gas enters the chamber tangentially, swirls through the chamber in a corkscrew motion, and exits. At the same time, liquid is sprayed inside the chamber. As the gas swirls around the chamber, pollutants are removed when they impact on liquid droplets, are thrown to the walls, and washed back down and out.



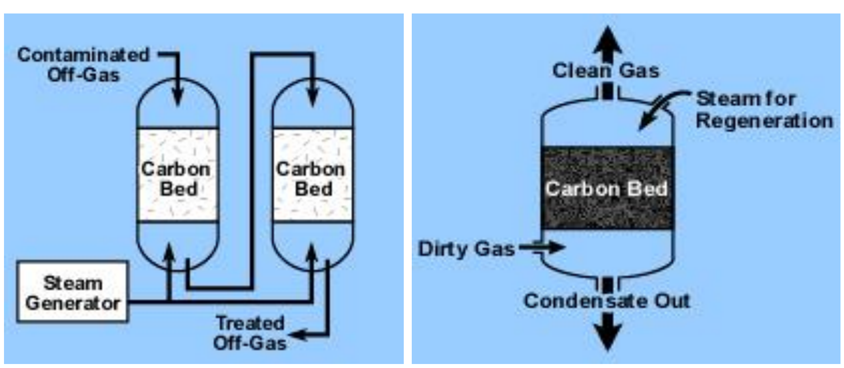
Venturi scrubber Packed tower scrubber Cyclone scrubber

II) Control devices for gaseous contaminants.

Adsorption:

Adsorption is the binding of molecules or particles to a surface. In this phenomenon molecules from a gas or liquid will be attached in a physical way to a surface. The binding to the surface is usually weak and reversible. The most common industrial adsorbents are activated carbon, silica gel, and alumina, because they have enormous surface areas per unit weight.

Carbon adsorption systems are either regenerative or non-regenerative. A regenerative system usually contains more than one carbon bed. As one bed actively removes pollutants, another bed is being regenerated for future use. Steam is used to purge captured pollutants from the bed to a pollutant recovery device. By "regenerating" the carbon bed, the same activated carbon particles can be used again and again. Regenerative systems are used when concentration of the pollutant in the gas stream is relatively high. Non-regenerative systems have thinner beds of activated carbon. In a non-regenerative absorber, the spent carbon is disposed of when it becomes saturated with the pollutant. Because of the solid waste problem generated by this type of system, non-regenerative carbon absorbers' are usually used when the pollutant concentration is extremely low.



C) Diffusion of pollution in air:

Dilution of the contaminants in the atmosphere can be accomplished through the use of tall stacks which penetrate the upper atmospheric layers and disperse the contaminants so that the ground level pollutants is greatly reduces. The height of the stacks is usually kept 2 to 2 1/2 times the height of nearby structures.

D) vegetation:

Plants contribute towards controlling air-pollution by utilizing carbon dioxide and releasing oxygen in the process of photosynthesis.

Plenty of trees should be planted especially around those areas which are declared as high-risk areas of pollution.

E) Zoning:

This method of controlling air pollution can be adopted at the planning stages of the city.

The Industrial estate of Bangalore is divided into three zones namely light, medium and large industries.

And other general methods for control of air pollution:

1. Source correction methods: use of quality raw materials, process changes, equipment modification or replacement etc.
2. Industrial estates should be established at a distance from residential areas.
3. Use of tall chimneys shall reduce pollution surroundings.
4. Poisonous gases reduced by passing the fumes through wet towers scrubbers or spray collectors.
5. Compulsory use of filters and electrostatic precipitators in chimneys.
6. Afforest ration.
7. Complete electrification of railway lines.
8. Use of pollution free fuels for vehicles. Ex. Alcohol, hydrogen, battery power etc.
9. Use non lead antiknock agent in gasoline.

4.3. WATER POLLUTION

The addition of any substances to water or changing of water physical and chemical characteristics in any way which interferes with its use for ultimate purposes is known as water pollution.

4.3.1. Causes of water pollution:

There are two major causes/ sources of water pollution, namely point sources and diffused sources.

i) Points sources: Those sources which can be identified at a single location are known as point sources. For instance, the flow of water pollutants through the regular channels like sewerage systems, industrial effluents etc. Infiltration of industrial effluents, municipal sewage etc. contaminates the ground water and cause water pollution.

Water pollution caused by point sources can be minimized if all domestic sewage, industrial effluents, cattle field and live stock waste water etc. are all centrally collected, treated up to requisite acceptable levels and reused for different beneficial purposes.

ii) Diffused sources: Those sources whose location cannot be easily identified are called diffused sources. In this case, the pollutants scattered on the ground ultimately reach the water sources and cause water pollution, for instance, agriculture (pesticides, fertilizers), mining, construction etc.

The water pollution caused by diffused sources like agriculture can be controlled by changing the cropping patterns, tillage practices and advanced farm management practices which do not contaminate the water bodies.

4.3.2. Water pollution and their sources:

Pollutant	**Principle sources**
Pathogenic organisms	Domestic waste (human and animal excreta), agricultural run off.
Solid wastes	Domestic, municipal, commercial, industrial and agricultural activities
Degradable organic matters -oil	Sewage, garbage, agricultural wastes, shipping accidents, run off from transport wastes, polluted land drainage, refineries wastes, off shore oil production.
Organic-pesticide (chlorinated hydrocarbons)	Application in agriculture, public health, industrial waste (pesticides, wood and carpet manufacturing)
Polychlorinated biphenyls	Sewage, industry (electrical, plastics, lubricants) and uncontrolled disposal of PCB containing products.
Detergents	Sewage and industrial wastes
Dyestuffs	Industrial wastes
Phenol	Industrial wastes
Inorganic acids and alkalis	Industrial wastes, burning of fossil fuels, medical and research laboratories, mercury industry (pulp, paper, mining, refining process), agricultural activities (seed treatment)
Lead	Anti-knocking ingredients of motor fuels, lead smelting chemical industry, lead paints, and enamels, agricultural pesticides.
Cadmium	Industry (mining and metallurgy, chemical, food industry, leather industry)
Arsenic	Industrial wastes, combustion of coal, food, agricultural use (fertilizers and insecticides)
Phosphates	Sewage, agricultural run off, industrial effluents

Nitrates and nitrites	Sewage, fossil fuel burning, industrial wastes and agricultural run off
Fluorides	Industrial processes (production of aluminium, steel, phosphate fertilizers, brick making) agricultural run off
Sulfides	Industrial effluents
Cyanides	Industrial effluents
Chlorates	Industrial effluents
CO_2	Carbon fuel combustion
SO_2	Industrial processes, burning of sulfur containing fuels
Ionizing radiation (including radio nuclides)	Medicinal uses, weapon production and testing, nuclear power production, use of radio isotopes and radiation
Heat	Fossil fuel and nuclear power stations, urban areas

4.3.3. Health hazards of water pollution:

a) Phosphrous and nitrates from fertilizers and detergents contaminate surface waters where they acts as nutrients and promote the growth of oxygen consuming algae which reduce the DO level of water, killing fish and other aquatic organisms.
b) Industrial effluents result in the addition of poisonous chemicals such as Arsenic, Mercury, Cadmium, Lead etc., which kill aquatic organisms and may reach human body through contaminated food (i.e. fishes etc)
c) Domestic, commercial and industrial effluents (petroleum refineries, paper mills, breweries, tanneries, slaughter houses) contaminate the water with organic pollutants. These provide nutrition for micro-organisms which decompose the organic matter and consume oxygen and reduce the DO level of the aquatic system thereby killing the aquatic organisms.
d) Non biodegradable pesticides (especially organochlorines) travel through food chains and ultimately reach humans where they accumulate in the fatty tissues and affect the nervous system.
e) Waterborne infectious enteric diseases like typhoid, bacillary dysentery, cholera and amoebic dysentery are the predominant health hazards arising from drinking contaminated water.
f) Fluorides containing pollutants cause fluorosis i.e. neuromuscular, respiratory, gastro intestinal and dental problems.
g) Thermal pollution of water reduces the DO level of the aquatic system making it incapable of supporting life.
h) Oil pollutants have been known to be responsible for the death of many water birds and fishes.
i) Radio-active pollutants (from mining and refining of Uranium, Thorium and Nuclear power plants) enter humans through food and water and get accumulated in the blood, thyroid gland, liver, bones and muscles.

4.3.4. Drinking Water quality standards:

INDIAN STANDARDS FOR DRINKING WATER
(BIS 10500 : 1991)

Sl. No.	Substance or Characteristic	Requirement (Desirable Limit)	Permissible Limit in the absence of Alternate source
Essential characteristics			
1.	Colour, (Hazen units, Max)	5	25
2.	Odour	Unobjectionable	Unobjectionable
3.	Taste	Agreeable	Agreeable
4.	Turbidity (NTU, Max)	5	10
5.	pH Value	6.5 to 8.5	No Relaxation
6.	Total Hardness (as $CaCo_3$) mg/lit., Max	300	600
7.	Iron (as Fe) mg/lit, Max	0.3	1.0
8.	Chlorides (as Cl) mg/lit, Max.	250	1000
9.	Residual, free chlorine, mg/lit, Min	0.2	--
Desirable Characteristics			
10.	Dissolved solids mg/lit, Max	500	2000
11.	Calcium (as Ca) mg/lit, Max	75	200
12.	Copper (as Cu) mg/lit, Max	0.05	1.5
13	Manganese (as Mn)mg/lit, Max	0.10	0.3
14	Sulfate (as SO_4) mg/lit, Max	200	400
15	Nitrate (as NO_3) mg/lit, Max	45	100
16	Fluoride (as F) mg/lit, Max	1.0	1.5
17	Phenolic compounds (as C_6H_5OH)mg/lit, Max.	0.001	0.002
18	Mercury (as Hg)mg/lit, Max	0.001	No relaxation
19	Cadmium (as Cd)mg/lit, Max	0.01	No relaxation
20	Selenium (as Se)mg/lit, Max	0.01	No relaxation
21	Arsenic (as As) mg/lit, Max	0.05	No relaxation
22	Cyanide (as CN) mg/lit, Max	0.05	No relaxation
23	Lead (as Pb) mg/lit, Max	0.05	No relaxation
24	Zinc (as Zn) mg/lit, Max	5	15
25	Chromium (as Cr^{6+})mg/lit, Max	0.05	No relaxation
26	Mineral Oil mg/lit, Max	0.01	0.03
27	Pesticides mg/l, Max	Absent	0.001
28	Alkalinity mg/lit.Max	200	600
29	Aluminium (as Al) mg/l,Max	0.03	0.2
30	Boron mg/lit,Max	1	5

4.3.5. Control measures for water pollution (Waste water Treatment):

Waste waters, whether domestic or industrial have several undesirable components- the organic and inorganic pollutants that are potentially harmful to the environment and human health. The treatment of waste water and its proper management has become a necessity in order to conserve this vital resource.

The main aim of waste water treatment is the removal of contaminants from water so that the treated water can be reused for beneficial purposes.

I. Various ways or techniques for control of water pollution are as follows:

a. Stabilization of ecosystem: This is the most scientific way to control water pollution. The basic principles involved are the reduction in waste input (thus control at source), harvesting and removal of biomass, trapping of nutrients, fish management and aeration.

b. Reutilization and recycling of waste: Urban waste (sewage) may be recycled to generate cheaper fuel gas and electricity. One distillery in Gujarat is able to treat 450000 liters of waste daily and generating energy equal to that produced by 10 tons of coal.

c. Removal of pollutants: Various pollutants (radioactive, chemical, biological) present in water body can be removed by appropriate methods such as adsorption, electrolysis, ion-exchange, reverse osmosis etc.

Ex: Ammonia removed by ion-exchange process, Mercury- selective ion exchange resin, phenolics- polymeric adsorbents, discoloration of water- electrolytic decomposition, sodium salts- reverse osmosis.

d. Setting up effluent treatment plants to treat waste.
e. Industrial wastes must be treated before discharge.
f. Educate public for preventing water pollution and the consequences of water pollution.
g. Strict enforcement of water pollution control act.
h. Continuous monitoring of water pollution at different places.
i. Developing economical method of water treatment.
j. River, streams, lakes and other water reservoirs must be well protected from being polluted.

II. Waste water treatment methods:

Wastewater treatment basically takes place in three stages:

1. Preliminary & Primary treatment, which removes 40-60% of the solids.
2. Secondary treatment, which removes about 90% of the pollutants and completes the process for the liquid portion of the separated wastewater.
3. Sludge (bio solids) treatment and disposal.

1. The primary treatment is of general nature and is used for removing suspended solids, odour, colour and to neutralize the high or low PH in the case of industrial effluents. This stage exploits the physical or chemical properties of the contaminants and removes the suspended and floating matter by screening, sedimentation, floatation, precipitation etc.

i)Screening: Screening devices are used to remove coarse solids from waste water.


Coarse Screen


Fine Screen


Perforated Plate Screen


Mesh Screen

ii) Sedimentation: In this step, the settle able solids are removed by gravitational setting under quiescent conditions. The sludge formed at the bottom of the tank is removed as under flow either by vaccum suction or by taking it to a discharge point at the bottom of the tank for withdrawal. The clear liquid produced is known as the overflow and it should contain no readily settle able matter.

iii) Floatation: Floatation may be used in place of sedimentation, primarily for treating industrial waste waters containing finely divided suspended solids and oily matter. Floatation technique is used in paper industry to recover fine fibers from the screened effluent and in the oil industry for the clarification of oil-bearing waste. It is also used for treating effluents from tanneries, metal finishing, cold-rolling and pharmaceutical industries.

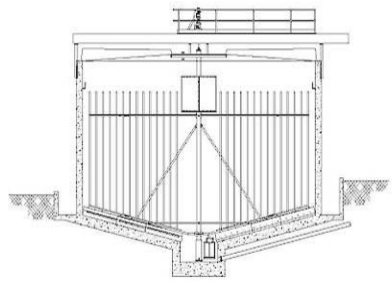
iv) Neutralization: When PH of the industrial waste is too high or too low then it should be neutralized by acid or alkali and only neutral effluent should be discharged into the drain or public sewer. For neutralization of acidic effluent, the following techniques are used- Limestone treatment, Caustic soda treatment etc.

2. Secondary or biological treatment: The biological process of sewage is a secondary treatment involving removing, stabilizing and rendering harmless very fine suspended matter, and solids of the waste water that remain even after the primary treatment has been done.

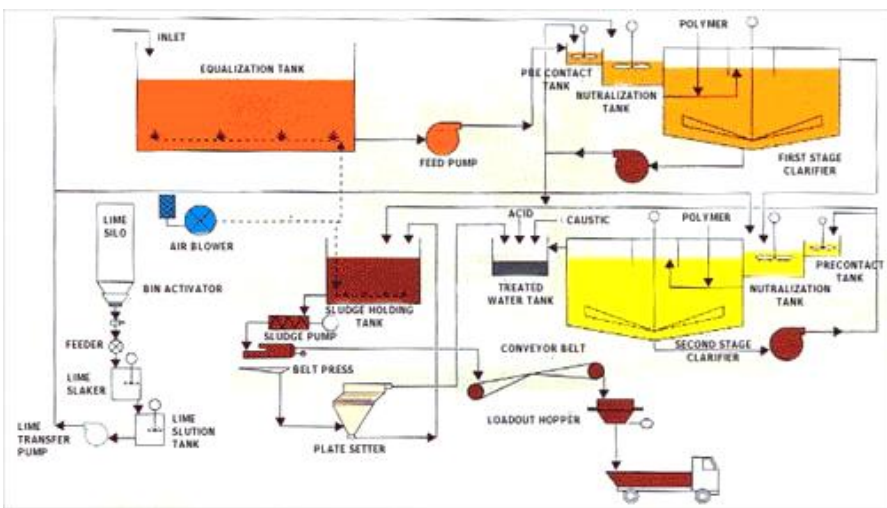
In biological treatment, oxygen supplied to the bacteria is consumed under controlled conditions so that most of the BOD is removed in the treatment plant rather than in the water course. Thus the principle requirements of a biological waste treatment process are an adequate amount of bacteria that feed on the organic material present in waste water, oxygen and some means of achieving contact between the bacteria and the organics.

Two of the most commonly used systems for biological waste treatment are the activated sludge system and biological film system.

a. Activated sludge system: In this system, the waste water is brought is brought into contact with a diverse group of microorganisms in the form of a flocculent suspension in an aerated tank.
b. In biological film system (trickling filters), the waste water if brought into contact with a mixed microbial population in the form of o film of slim attached to the surface of a solid support system. In both cases the organic matter is metabolized to more stable inorganic forms.



3. Tertiary or advanced waste water treatment: Usually the primary and secondary treatments are sufficient to meet waste water effluent standards. However, if water produced is required to be of higher water quality standards (in case the water to be put to some direct reuse) the advanced waste water treatment is carried out. More than one tertiary treatment process may be used at any treatment plant. If disinfection is practiced, it is always the final process. It is also called "effluent polishing." Advanced methods are Ammonia removed by ion-exchange process, Mercury- selective ion exchange resin, phenolics- polymeric adsorbents, discoloration of water- electrolytic decomposition, sodium salts- reverse osmosis.

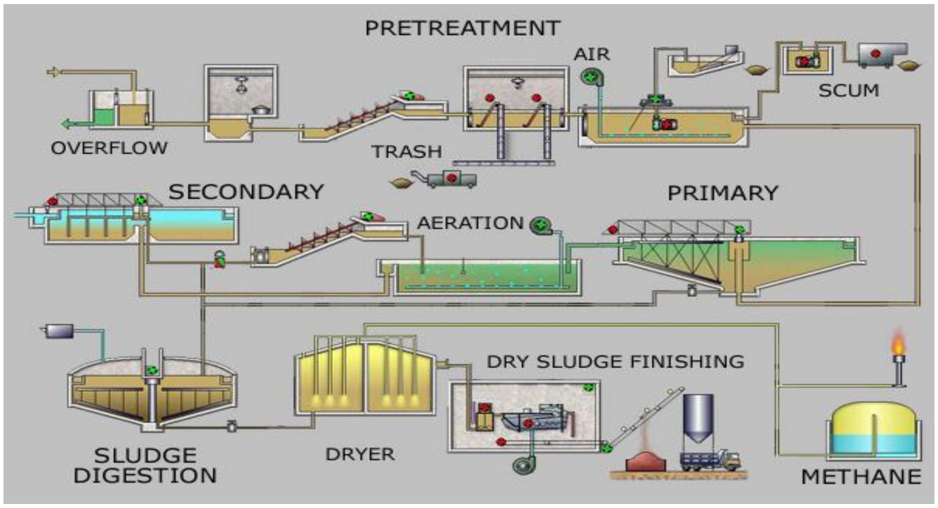


Effluent Treatment Plant Schematic Diagram

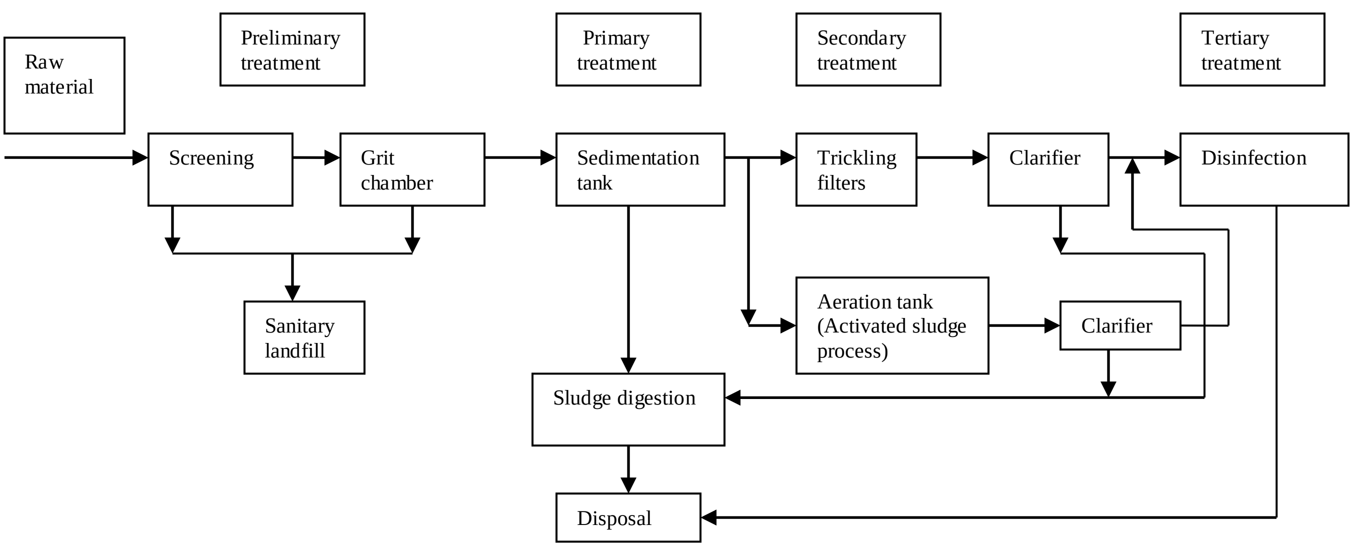
Sewage treatment plant:

Sewage can be treated close to where it is created (in septic tanks, bio filters or aerobic treatment systems), or collected and transported via a network of pipes and pump stations to a municipal treatment plant. Sewage collection and treatment is typically subject to local, state and federal regulations and standards. Industrial sources of wastewater often require specialized treatment processes. Conventional sewage treatment may involve three stages, called primary, secondary and tertiary treatment.

Process Flow Diagram for a typical large-scale treatment plant



Layout of Municipal Sewage Treatment Plant:



Common /Combined Effluent Treatment Plant (CETP)

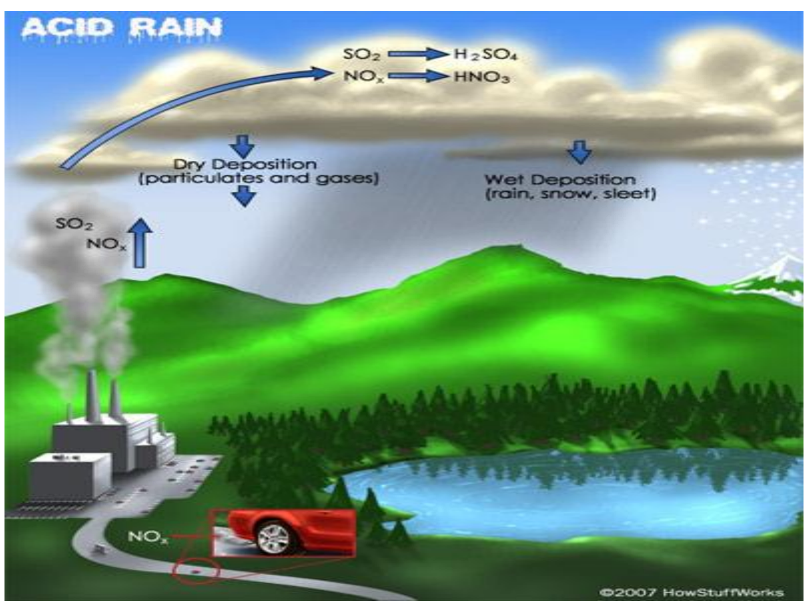
Common Effluent Treatment Plant is the concept of treating effluents by means of a collective effort mainly for a cluster of small scale industrial units. This concept is similar to the concept of Municipal Corporation treating sewage of all the individual houses. The main objective of CETP is to reduce the treatment cost for individual units while protecting the environment.

- To achieve 'Economics of scale' in waste treatment, thereby reducing the cost of pollution abatement for individual factory.
- To minimize the problem of lack of technical assistance and trained personnel as fewer plants require fewer people.
- To solve the problem of lack of space as the centralized facility can be planned in advance to ensure that adequate space is available.
- To reduce the problems of monitoring for the pollution control boards.

4.4. SOIL POLLUTION

In modern economics, various types of activity, including agriculture, industry and transportation, produce a large amount of wastes and new types of pollutants.

4.4.1. Soil as a sink for pollutants:



1. Chemical pollutants: The chemical pollutants such as calcium carbonate, bicarbonates, calcium sulphate, and soluble salts etc. from eroded sediments pollute the soil. It is estimated that 85% of phosphorus and about 70% of nitrogen loading of surface water are brought from eroded soil from hills or other places. The tanneries, synthetic drug factories and distilleries discharge lot of suspended and dissolved solids which pollute the

soil. The presence of all these substances also retards growth of plants, retards reproduction process and also fruit production.

2. Metallic pollution: The metallic pollutants from Cu, Steel, Cd, Zn, factories pollute the soil due to excess of Cu, Fe, Cd and Zn. Besides this, the presence of Co, Ni, Pb, Ba, Mn, Al, Sr, Silicon, Ca etc, added to the soil from various industries in combined form also pollute the soil such pollution by metals is called metallic pollution.

3. Industrial effluents: Industries are major sources of soil pollution now a days. Sugar factories, textile, steel, paper, chemical and pesticide, petroleum, engineering, cement, glass, dyeing, oil refineries and other factories are responsible for addition of more than 40 million tones of substances in the soil as industrial wastes.

These toxic chemicals through soil and water enter into vegetables, fruits, grains, etc. and enter into food chain of human beings and are responsible for number of diseases and even cancer.

4. Agricultural wastes: For increasing the production in agriculture – lot of fertilizers, pesticides, herbicides etc. are added in each season with the soil is getting harder every year due to these inorganic chemicals called agricultural wastes in agricultural fields.

Soil conditioners, fumigants contain toxic metals such as Cd, Hg, Co, Pb, etc. increase of their concentration in soil they enter into crops and finally to food chain of human being causing mental, skin, lung, blood and urine diseases.

Today 30% diseases in human beings are suffered due to presence of fungicides, insecticides, herbicides etc, Fertilizers add phosphorus, nitrogen, sodium, potassium, sulphate, nitrate etc. in the soil. The high concentration of nitrates and phosphates also cause eutrophication, choking the whole aquatic ecosystem in nature.

The animal waste have been found to contain high BOD (>300 ppm) and high COD (>500 ppm) and nitrogen (>450 ppm) and hence such wastes are very harmful for human beings.

The villages in India, villagers use cow dung for burning purposes but it is dangerous for health as it generates benzo-pyrene in smoke which causes cancer to human beings.

5. Urban wastes: The urban wastes contain substances like glass bottles, glass plates, plastics, paper, rubber, fibers, iron pieces, garden leaves, branches, flowers, parts of vehicles, food residues, vegetable residues, fuel residues, dust, metallic pieces, cane etc. polythene bags and other household articles.

6. Radio active pollutants: The radio nuclides such as Iodine -129, Caesium -137, Barium -140, Strantium – 90, Promothium- 144, Ruthenium- 106 which are produced from nuclear fission get deposited on the soil which continuously emit gamma radiation which are harmful for plants, aquatic life and human beings.

7. Biological gents: The biological agents cause pollution of soils and crops. Bacteria, algae, nematodes, actinomycetes, rotifers, protozoans etc. are important biological agents which change the physical texture of the soil and are responsible for changing the fertility of the soil. The human and animals excreta are the major source of land pollution.

8. Pesticides: Pesticides like DDT, BHC, malathion, endrin, aldrin, parathion etc. cause impairment of human tissues, failure in the functioning of liver, kidney, intensive and gonads.

9. Detergents: Detergents pollute the soil and river water. They destroy the fertility of the soil and retard the growth of plants and fruits.

4.4.2. Impact of modern agriculture on soil:

1. Fertilizer application promoting the crop yield will deplete the micro nutrients in the soil, it progressively infertile.
2. Soil micro flora loses.

3. Soil Salinity:

Accumulation of excessive salts in the soil system makes it unfit for cultivation. Accumulation of excessive salts in soil is known as 'Salt efflorescence'.

Causes:

- Water logging is the main cause of salt efflorescence. Continuous evaporation from water logged areas leaving the dissolved salts behind, causes accumulation of excessive salts.
- Excessive application of chemical fertilizers can increase the soil salt content.
- Percolation of domestic and industrial sewage adds up to the soil salts.

Effects of soil salinity:

1. When the salinity increases the tolerance limit of a crop, the land is unfit for the crop.
2. Decreasing of plant yield.
3. Percolation of rain water the ground water turns saline by the lechates of the soil with excessive salts.
4. Soil bacterial system are upset. Soil bacteria play a vital role in the equilibrium of an ecosystem in nature.

4. Water logging:

Water logging is the phenomenon of ground water table rising too close to the ground level to sustain useful plant life.

Causes of water logging:

1. Over irrigation of land
2. Seepage losses from canals
3. Surface spreading of waste waters
4. Excessive rainfall
5. Poor land drainage .
6. Poor permeability of the soil system etc.

Effects of water logging:

1. The land cannot sustain useful plant life.
2. Evaporation from water logged areas leaving salts behind, builds up the soil salinity to a level that inhibits vegetation.
3. Mosquitoes and other vectors and worms proliferate.
4. Organic matter in the soil under goes an anaerobic decomposition due to water logging, creating bad odors and ground water pollution.
5. Water supply systems and waste water disposal systems face additional difficulties in design and operation.
6. Structural foundations are difficult to construct.

4.4.3. Degradation of soil:

Land degradation refers to deterioration or loss of fertility or productive capacity of the soil. Land resources are very much related to natural disasters like volcanic eruptions, earthquakes etc. but it is due to human activities that soil gets polluted. Land degraded by soil erosion, salination, water-logging, desertification, shifting cultivation, urbanization, landslides and soil pollution.

A. Soil erosion:

The top layer of the soil contains nutrient required by the plants. This fertile top soil is most valuable natural resource, at a depth of 15 – 20 cm. The loss of top soil or disturbance of the soil structure is known as 'soil erosion'.

Types of soil erosion:

Based on the rate of soil loss takes place, there are two main types-

1. Normal or geologic erosion: It occurs under normal, natural conditions by it-self without any interference of man. As natural, it is very slow process, unless there is some major disturbance by a foreign agent.
2. Accelerated soil erosion: This type of removal of soil is very rapid and never keeps place with the soil formation. This is the serious type of loss, generally caused by an interference of an agency like man and other animals.

Agents of soil erosion:

Based on the various 'agents' that bring about soil erosion and the 'form' in which the soil is lost during erosion, there are following types-

a. Water erosion: caused by the action of water, which removes the soil by falling on as 'rain drops', as well as by its 'surface flow' action.
b. Wind erosion: Soil loss by wind is common in dry (arid) regions, where soil is chiefly sandy and the vegetation is very poor or even absent. High velocity winds blow away the soil particles.
c. Landslides or Slip erosion: The hydraulic pressure caused by heavy rains increases the weight of the rocks at cliffs which come under gravitational force and finally slip or fall off. Sometimes entire hillock may slide down.
d. Stream bank erosion: The river during floods splash their water against the banks and thus cuts through them.
e. Deforestation & Over grazing: Effects of deforestation and over grazing are well known on soil loss. In India annual loss of soil nutrients in this way is of the order of 5.37 million tons of N, P, K valued at about Rs. 700 crores.

B. Salination: refers to increase in the concentration of soluble salts in the soil. Poor drainage of irrigation and flood waters results in accumulation of dissolved salts on the soil surface. In arid and semi-arid areas with poor drainage and high temperatures, water evaporates quickly leaving behind a white crust of salts on the soil surface. The high concentration of salts in soil severely affects the water absorption process of the plants, resulting into poor productivity.

C. Water logging: Water logging may be due to surface flooding or due to high water table. Excessive use of canal irrigation may disturb the water balance and create water logging as a result of seepage or rise in the water table of the area. The productivity of water logged soil is severely affected reduce due to lesser availability of oxygen for the respiration of plants.

D. Desertification: Desertification is a slow process of land degradation that leads to desert formation. I is like a 'skin disease' over planet wherein patches of degraded land, eruption separately, gradually join together. For example, Thar desert (India) was formed by the degradation of thousands of hectares of productive land. It may result either due to a natural phenomenon linked to climatic change or due to abusive use of land. Mismanagement of natural resources including land, certain irreversible changes have triggered the break down of nutrient cycles and micro climatic equilibrium in the soil indicating the ones of desert conditions.
Causes of desertification:
a. Deforestation: The process of denuding and degrading a forested land initiates a desert producing cycle that feeds on itself. Since there is no vegetation to hold back the surface run-off, water drains off quickly before it can soak into the soil to nourish the plants or to replenish the groundwater. This increases soil erosion, loss of fertility and loss of water.
b. Over grazing: This is because the increasing cattle population heavily graze in grasslands or forests and as a result denude the land area. The dry barren land becomes loose and more prone to soil erosion. The top fertile layer is also lost and thus plant growth is badly hampered in such soils.
c. Mining and quarrying: These activities are also responsible for loss of vegetative cover and denudation of extensive land areas leading to desertification.
Amongst the most badly affected areas are the sub Saharan Africa, the Middle East, Western Asia, parts of Central and South America, Australia and the Western half of the US.

E. Shifting cultivation: Shifting (Jhum) cultivation, a very peculiar practice of slash and burn agriculture, prevalent among many tribal communities inhabiting the tropical and sub- tropical regions of Africa, Asia and Islands of Pacific Ocean has also laid large forest tracts bare. This practice has led to complete destruction of forests in many hilly areas of India, North-East and Orissa, and caused soil erosion and other associated problems of land degradation.

F. Urbanization: Human activities are responsible for the land degradation of forests, croplands and grass lands. The productive areas are fast reducing because of urbanization i.e., the developmental activities such as human settlements and industries.

G. Landslides: Human activities such as construction of road and railway, canal, dams and reservoir and mining in hilly areas have affected the stability of hill slopes and damaged the protective vegetation cover above and below roads and other such developmental works. This has upset the balance of nature, making such areas vulnerable to landslides.

H. Soil pollution: Industrial effluents contain high concentration of chemicals. For example Paper and Pulp industrial waste water contain organic matter, Soap industrial waste water contain fatty acids, fertilizer industrial waste water contain chemicals etc. these chemicals degrade the soil fertility and loss of micro organisms.

4.4.4. Control of soil pollution:

1. The garbage can be converted into compost manure by cobalt and nickel techniques recently developed.
2. Half of the total dung collected in villages is used as fuel. This must be stopped at once as this fuel on burning releases benzo-pyrene which cause cancer. If this bio-manure is used in fields the crop production will be increased.
3. Solid wastes should be treated by pyrolysis method. Fluided bed furnaces and multiple hearth furnaces can also be used to yield better results.
4. Salts flow to soil from industries must be stopped.
5. Environment protection Act, 1986 be implemented with full force to factories and municipal corporations to check the flow of untreated wastes directly into the soils.
6. The land act should be implemented strictly against the industries which pollute the soil.
7. A huge plantation is necessary along the roads, fields, waste places, marshes, and nearby railway linings. Among these plants planted on the boundary of the fields to avoid soil erosion and to increase the fertility of the soil.
8. The industrial effluents should be first treated and then allowed to discharge through constructed drains.
9. The use of pesticides should be avoided as it will destroy the fertility of the soil in time to come.
10. The organic manure be used in the fields in place of chemical fertilizers as later will also destroy gradually the fertility of the soil.
11. The garbage first be treated and then it should be converted into manure as is happening in western countries. It should never be used in fields as it will destroy the fertility of the soil permanently.
12. Environmental awareness programs be arranged to create awareness among students and others to check the pollution of soil.
13. Toxic and non-degradable materials must be totally banned.
14. Recycling and reuse of industrial and domestic wastes can minimize soil pollution considerably.

4.5. Marine Pollution or Sea pollution

Marine pollution is defined as discharge of substances to the marine environment resulting in adverse effects such as hazards to human health, obstruction in marine activities, marine water and coastal land zones.
According to a rough estimate, about
Lead (Pb) : 12,000 tons, Copper (Cu): 17,000 tons, Zinc (Zn): 70,000 tons,
Arsenic (As): 8,000 tons, Barium (Ba): 900 tons, Manganese (Mn): 70,000 tons,
Chromium (Cr): 4,600 tons, Antimony (An): 3,800 tons, Iron (Fe): 17,000 tons
Mercury (Hg): 7,000 tons
Tin (Sn): 4,600 tons are discharged per year into the sea without any dilution. The rapid industrialization has caused havoc for sea life and posed a serious threat to fishery resources. The colored dyes and other chemicals have changed the color of the sea at certain places and transparency has completely vanished.
Once in 1985 in Haji port of Bombay, about 80,000 fishes were found dead due to industrial effluents containing cyanide irons and mercury. Due to mercury poisoned fishes, about 20,000 people in Japan fell sick and many died in 1978. The disease was given the name Minamata on the name of the city of Japan.
Almost all rivers of the world are highly polluted and they carry the domestic sewage and mixture of industrial effluents into the sea without any treatment.

4.5.1. Misuse of international water for dumping of hazardous water:

Main sources of marine pollution:

1. Municipal waste and sewage from residues and hotels in coastal cities are directly discharged into sea.
2. Dumping of industrial effluents e.g. acids, pesticides etc.
3. Oil spills by oil refinery companies (oil exploration).
4. Wastes of ship yards and shipping companies.
5. Solid wastes- garbage, trash, explosive, radio active waste.

Ocean dumping: is the practice of dumping all sorts of garbage in the oceanic waters. Industrial waste reigns supreme in this process, since the oceans provide a convenient out-of-sight, out-of-mind way to get rid of the junk that industries churn out during the course of production. But industries aren’t the only culprits. Apart from the construction waste, demolition debris contributed by them, there is also a whole lot of other stuff. Sewage, chemical and nuclear waste of research foundations and the all-pervasive littering of beaches and other coastal picnic and recreational areas also add to the ocean trash.
A report by the United Nations Environmental Programme (UNEP) says that plastic, especially plastic bags and polyethylene tere phthalate (PET) is the most common of all ocean pollutants. Also, plastic is the most dangerous to be dumped into the ocean with regards to marine life.

There are three main direct public health risks from ocean dumping: the first one being the risk to humans by the consumption of contaminated marine organisms, the other occupational hazards of injuries and accidents faced by the fishermen whose livelihood depends upon the state of the oceans and the exposure of humans to the waste that is washed up on the beaches. Periodically, medical and other wastes from both legal and illegal dumping have washed up on beaches, resulting in exposure to beachgoers and, in some cases, the closure of beaches until the wastes could be removed. Consumption of fish contaminated from radioactive wastes may pose a serious problem worldwide because of nuclear waste dumping in the oceans.

4.5.2. Coastal pollution due to sewage and marine disposal of industrial effluents:

Coastal waters receive a variety of land-based water pollutants, ranging from petroleum wastes to pesticides to excess sediments. Marine waters also receive wastes directly from offshore activities, such as ocean-based dumping (e.g., from ships and offshore oil and gas operations).

One pollutant in the ocean is sewage. Human sewage largely consists of excrement from toilet-flushing; wastewater from bathing, laundry, and dishwashing; and animal and vegetable matter from food preparation that is disposed through an in-sink garbage disposal. Because coasts are densely populated, the amount of sewage reaching seas and oceans is of particular concern because some substances it contains can harm **ecosystems** and pose a significant public health threat. Animal wastes from feedlots and other agricultural operations (e.g., manure- spreading on cropland) pose concerns similar to those of human wastes by virtue of their microbial composition.
The major types of ocean pollutants from industrial sources can be generally categorized as petroleum, hazardous, thermal, and radioactive. Petroleum products are oil and oil-derived chemicals used for fuel, manufacturing, plastics-making, and many other purposes. Hazardous wastes are chemicals that are toxic (poisonous at certain levels), reactive (capable of producing explosive gases), corrosive (able to corrode steel), or ignitable (flammable). Thermal wastes are heated wastewaters, typically from power plants and factories, where water is used for cooling purposes. Radioactive wastes contain chemical elements having an unstable nucleus that will spontaneously decay with the concurrent emission of ionizing radiation.

4.5.3. Effects of Marine pollution:

1. Death of retarded growth, vitality and reproductively of marine organisms cause by toxic pollutants.
2. Reduction in the dissolved oxygen content necessary for marine life because of increased biochemical oxygen demand (BOD) as a result of the decomposition of organic wastes.
3. Because of nutrient rich waste, depletion of oxygen and subsequent killing of algae that may wash up and pollute coastal areas.
4. Waste disposal causes habitat change that drastically change entire marine

ecosystem.
5. Contaminated marine organisms may transmit toxic elements or diseases to people who eat them
6. Beaches have been closed (temporarily) to recreational uses.

4.5.4. Control of Marine pollution:

i) Control of oil pollution :

Physical and chemical methods:

1. Absorbents: Oil spill can be cleaned by using absorbents such as moss, saw peat, dust, pine bark and straw etc. If the absorbent material is removed from the water, the oil can be removed easily and used.
2. Emulsification: It is simple process in which water is incorporated into the floating oil forming a water- in- oil emulsion. Such emulsions, containing 20 to 80% water are often viscous and are called “mousse”.
3. Using chemical additives: Chemical additives are generally used to solidify oil from water surface. Ex. Detergents.
4. Mechanical methods are also used to remove oil slicks from marine water.
5. Improved navigation Aids: The offshore drilling operations can be effectively protect the water from oil pollution.
6. Role of micro organisms in oil clean-up in the ocean: In the clean up operations either genetically engineered organisms or biotechnological approaches have been adopted to solve oil pollution problems. Certain bacteria like “Pseudomonas” are also being employed in the recovery of oil.

ii) Other chemical pollutants:

1. Municipal and domestic wastes should be treated before being allowed to mix with sea.
2. Periodically analysis of coaster water.
3. Oil spills in marine should be rectified by means of bio remediation.
4. Beaches should be maintained in neat and tidy manner.
5. Discharge of nutrients from sewage can promote growth of algae hence sewage should be recycled before discharge.
6. Educate people regarding marine ecosystems.

4.6. Noise Pollution

“Unwanted sound dumped into the atmosphere leading to health hazards”.
Or “Wrong sound in the wrong place at the wrong time”.

4.6.1. Sources:
Main contributors to noise are industries, transportation (air, rail and road) and community and religious activities.

Man made sources in urban areas are automobiles, industries, trains, air plans, noise makers are horns, sirens, lawn movers, T.V., radio, transistors, telephone, dogs, loud speakers, washing machines etc.

Properties:

Just audible sound is about 10db, library place 30db, heavy street traffic 60-80db, boiler factories 120db, jet planes (take off)- 150db, rocket engines 180db.

4.6.2. Industrial noise is usually considered mainly from the point of view of environmental health and safety, rather than nuisance, as sustained exposure can cause permanent hearing damage. Traditionally, workplace noise has been a hazard linked to heavy industries such as ship-building and associated only with noise induced hearing loss (NIHL). Modern thinking in occupational safety and health identifies noise as hazardous to worker safety and health in many places of employment and by a variety of means.

Noise can not only cause hearing impairment(at long-term exposures of over 85 decibels (dB), known as an exposure action value), but it also acts as a causal factor for stress and raises systolic blood pressure.

Additionally, it can be a causal factor in work accidents, both by masking hazards and warning signals, and by impeding concentration.

Noise also acts synergistically with other hazards to increase the risk of harm to workers. In particular, noise and dangerous substances (e.g. some solvents) that have some tendencies towards ototoxicity (is damage to the ear (*oto*-), specifically the cochlea or auditory nerve and sometimes the vestibular system, by a toxin.) may give rise to rapid ear damage.

4.6.3. Occupational health hazards:

Effects are categorized as 1. auditory effects (affecting on hearing faculty) and 2. non-auditory effects (other than auditory ones).

1. Auditory effects: These include “auditory fatigue” appears in the 90db and may be associated with side effects as whistling and buzzing in ears.
   “Deafness” caused due to continuous noise exposure. Temporary deafness occurs at 40 – 60 db noise. Permanent loss of hearing occurs at 100db. Bombay and Calcutta are the nosiest cities in the world.

2. Non- auditory effects: These are
   a. Interference with speech communication, noise of 50-60db.
   b. Annoyance (irritation): balanced persons express great annoyance at even low level of noise as crowd, highway, radio etc. Effects are in temper, bickering etc.
   c. Loss of working efficiency.
   d. Physiological disorder: neurosis, anxiety, hypertension, behavioral and emotional stress, increase of swatting etc.

4.6.4. Standards for noise pollution:

SCHEDULE

see rule 3(l) and 4(l)

Ambient Air Quality Standards in respect of Noise

Area Code	Category of Area/Zone	Limits in dB(A) Leq *	
		Day Time	NightTime
(A)	Industrial area	75	70
(B)	Commercial area	65	55
(C)	Residential area	55	45
(D)	Silence Zone	50	40

Noise Limit for Vehicles

Category of Vehicles	Maximum Permissible Noise Level
Two wheelers (Petrol driven)	80 dB (A)
All passengers cars, all Petrol driven three- wheelers and diesel driven two wheelers	82 dB (A)
Passenger or Light Commercial Vehicles including three wheelers vehicles fitted with diesel engine with gross vehicles weight up to 4000 kgs.	85 dB (A)
Passenger or Commercial Vehicles with gross vehicles weight above 4000 kgs and up to 12000 kgs	89 dB (A)
Passenger or Commercial Vehicles with gross vehicles weight above 12000 kgs.	91 dB (A)

Noise standards for fire-crackers

(i) The manufacture, sale or use of firecrackers generating noise level exceeding 125 dB(AI) or 145 dB(C) pk at 4 meters distance from the point of bursting shall be prohibited.

(ii) For individual fire-cracker constituting the series (joined firecrackers), the above mentioned limit be reduced by 5 log 10 (N) dB, where N = number of crackers joined together.

The broad requirements for measurement of noise from firecrackers shall be-

The measurements shall be made on a hard concrete surface of minimum 5 meter diameter or equivalent.

The measurements shall be made in free field conditions i.e., there shall not be any reflecting surface upto 15 meter distance from the point of bursting.
The measurement shall be made with an approved sound level meter.
The Department of Explosives shall ensure implementation of these standards.
The firecrackers for the purpose of export shall be exempted from the sub-paragraphs A, B and C above, subject to the compliance of the following conditions, namely:- the manufacturer shall have an export order;
the firecrackers shall conform to the level prescribed in the country to which it is exported; they shall have a different packing colour code, and there shall be a declaration on the box “not for sale in India ” or “only for export in other countries.”
Note: dB(AI) : A-weighted impulse Sound Pressure Level in decibel
dB(C) pk : C-weighted Peak Sound Pressure Level in decibel”

Noise Limit for Generator Sets run with Petrol or Kerosene

	Noise Limit from	
	September 1, 2002	**September 1, 2003**
Sound Power Level L_{WA}	90 dBA	86 dBA

4.6.5. Control of noise pollution:

Some towns and cities actually have laws and limits as to the noise pollution that its citizens can make. Other places, like golf courses or nature reserves, are also considered to be "quieter" places with less noise pollution. There are some things that all people can do to help lower their own levels of noise pollution that they - and hopefully others around them - produce.

1. Control at source: This can be done by- designing and fabricating silencing devices in air crafts engines, automobiles, industrial machinery and home appliances. And by segregating the noisy machines.
2. Transmission control: This can be done by- Covering the room walls with sound absorbers as acoustic tiles and construction of enclosures around industrial machinery.
3. To protect exposed person: The workers exposed to noise can be provided with wearing devices as ear plugs and ear muffs.
4. To create vegetation cover: Trees should be planted along highways, streets and other places. Ashok, Neem, Tamarind etc trees are good for this purpose.
5. Noise control through law: Noise control act implement with full force against noise generating industries, automobiles and public.
6. Education: Public must be made aware and educated about noise nuisance through adequate news media, lectures and other programs.

<u>Other measures:</u>

- Rake leaves by hand; don't use a noisy leaf blower.
- Trim bushes or shrubs by hand; don't use a noisy bush trimmer.

- Sound proof rooms that might have music conducted in them, like a room with a piano or if someone in the house plays drums or guitar or whatever. This can be done simply with curtains, window inserts, carpeting, and closing windows and doors.
- Don't blast music on the radio or computer or speakers. Be considerate of your own ears and those of other around you.
- Don't slam doors / car doors, close them early and with only as much force as needed. (People don't usually think of this, but imagine - how loud is it when you slam your car door? Pretty tolerable. But imagine thousands of people doing so. Now *that* can start to get loud.
- Turn off the TV or radio when you aren't actually fully listening to it.
- Train your dog not to bark so much.
- Don't yell. Have civil conversations. Call someone or go find them instead of yelling across the street for them, for example.
- Plant trees and bushes around you house. They help keep the air clean, absorb sound, give privacy, and add nice design and looks to a house.
- Do noisy things (dishes, hammering, etc.) over or on a rubber mat to reduce noise.
- Put carpets, rugs, mats, throw rugs, etc. in your house / mats outside.
- Put fabric window coverings instead of plastic or wooden shades / blinds.
- Don't rev up a motorcycle or car unless it is actually needed for the drive.
- Don't beep your car horn "just because", make sure it is a legitimate reason.

4.7. Thermal Pollution

4.7.1. Thermal comfort:

Human **thermal comfort** is defined as the state of mind that expresses satisfaction with the surrounding environment. Thermal comfort is affected by heat conduction, convection, radiation, and evaporative heat loss. Thermal comfort is maintained when the heat generated by human metabolism is allowed to dissipate, thus maintaining thermal equilibrium with the surroundings. Any heat gain or loss beyond this generates a sensation of discomfort. It has been long recognized that the sensation of feeling hot or cold is not just dependent on air temperature alone.

Thermal comfort is very important to many work-related factors. It can affect the distraction levels of the workers, and in turn affect their performance and productivity of their work. Also, thermal discomfort has been known to lead to Sick Building Syndrome symptoms.

The occurrence of symptoms increased much more with raised indoor temperatures in the winter than in the summer due to the larger difference created between indoor and outdoor temperatures.

4.7.2. Heat Island effect:

The term "heat island" describes built up areas that are hotter than nearby rural areas. The annual mean air temperature of a city with 1 million people or more can be 1.8–5.4°F (1–3°C) warmer than its surroundings. In the evening, the difference can be as high as 22°F (12°C). Heat islands can affect communities by increasing summertime peak energy demand, air conditioning costs, air pollution and greenhouse gas emissions, heat-related illness and mortality, and water quality.

Elevated temperatures from urban heat islands, particularly during the summer, can affect a community’s environment and quality of life. While some impacts may be beneficial, such as lengthening the plant-growing season, the majority of them are negative. These impacts include:

- Increased Energy Consumption
- Elevated Emissions of Air Pollutants and Greenhouse Gases
- Compromised Human Health and Comfort
- Impaired Water Quality

4.7.2. Solar Radiation effects:

i) Effects on human health: The body produces vitamin D from sunlight (specifically from the UVB band of ultraviolet light), and excessive seclusion from the sun can lead to deficiency unless adequate amounts are obtained through diet. This very common deficiency leaves the body generally vulnerable to cancers.

On the other hand, excessive sunlight exposure has been linked to all types of skin cancer caused by the ultraviolet part of radiation from sunlight or sunlamps. Sunburn can have mild to severe inflammation effects on skin; this can be avoided by using a proper sunscreen cream or lotion or by gradually building up melanocytes with increasing exposure. Another detrimental effect of UV exposure is accelerated skin aging (also called skin photo damage), which produces a difficult to treat cosmetic effect. Some people are concerned that ozone depletion is increasing the incidence of such health hazards. A 10% decrease in ozone could cause a 25% increase in skin cancer.

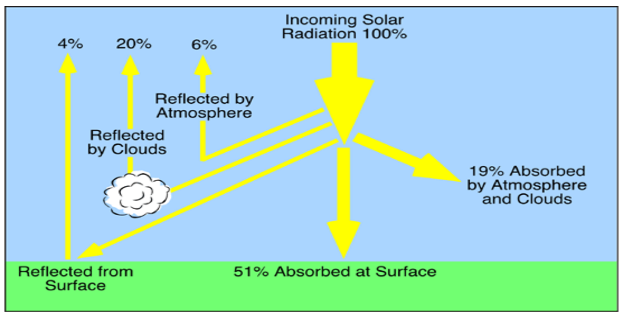
ii) Effect of sun angle on climate: The amount of heat energy received at any location on the globe is a direct effect of sun angle on climate, as the angle at which sunlight strikes the Earth varies by location, time of day, and season due to the Earth's orbit around the sun and the Earth's rotation around its tilted axis.

When sunlight shines on the earth at a lower angle (sun closer to the horizon), the energy of the sunlight is spread over a larger area, and is therefore weaker than if the sun is higher overhead and the energy is concentrated on a smaller area.

The sunbeam hitting the ground at a 30° angle spreads the same amount of light over twice as much area (if we imagine the sun shining from the south at noon, the north-south width doubles; the east-west width does not). Consequently, the amount of light falling on each square mile is only half as much.

The sunbeam entering at the shallower angle must also travel twice as far through the Earth's atmosphere, which reflects some of the energy back into space.

The angle of sunlight striking the earth in the Northern and Southern hemispheres when the Earth's northern axis is tilted away from the sun, when it is winter in the north and summer in the south.



4.8. Solid Waste Management

4.8.1. Solid waste classification:

Solid wastes may be classified based on content and partly on moisture and heating value. That are-

1. Garbage: These waste produced during the preparation or storage of meat, fruit, vegetables etc.
2. Rubbish: In this combustible wastes would include paper, wood, scrap, rubber, leather etc. Non combustible wastes are metals, glass, ceramics etc.
3. Pathological wastes: Dead animals, human wastes etc.
4. Industrial wastes: Chemicals, paints, sand, metal ore processing, fly ash, sewage treatment sludge etc.
5. Agricultural wastes: Farm animal manure, crop residues etc.

On the basis of nature of solid wastes, it can be categorized into two types.

1. Urban wastes: It is different types-

a. Domestic wastes ex: Food, cloth, paper, glass, plastic.

b. Commercial wastes ex: Packing material, cans, bottles, plastic, rubber.

c. Construction wastes ex: Wood, concrete, debris, lime, cement.

d. Biomedical wastes ex: Anatomical wastes, infected wastes.

2. Industrial wastes:

a. Nuclear power plants ex: radio active wastes
b. Thermal power plants ex: ash, water, un burnt fuel.
c. Chemical industries ex: toxic chemicals, oxides, acids.
d. Other industries ex. Wood scrap, oil, paint, dyes, lime, cement.
Non - hazardous solid wastes from city, town or village is called municipal solid waste (MSW). Municipal solid waste includes domestic wastes, commercial wastes and biomedical wastes.

4.8.2. Collection of solid waste:

Municipal solid waste:

1. Solid Waste Sorting:

Solid waste can have lots of different descriptions. Yard waste such as grass clippings and unused mulch is technically solid waste. So is regular trash that a person may just throw away like spoiled food or paper plates. Recyclables such as cardboard and newspaper are also considered solid waste. The first step in the collection process is to sort the waste by type and define what will and won't be disposed of. Once the waste has been sorted, usually by the residents putting it in the proper bin or container, the next step in collection can take place.

2. Pickup:

There are a variety of pickup options. Trash can be placed on the curb near a home, and garbage collectors can go house to house and take it. There are also communal trash bins in some apartment complexes, as well as county recycling containers for glass, plastic, paper and organic waste. Some sites may collect hazardous waste materials like syringes, which can't be left in regular trash sites. The waste is taken from all of these sites on a schedule and then transported to a final destination.

There are two types of containers for collection of solid waste:

i) Stationary container system: In this method solid waste is collected from bin in the centers and collected solid waste transport to disposal sites by tractor or lorry.
ii) Hawl container system: In this filled containers are replaced by empty containers. These filled bins are transport to disposal site. After dumping the empty bins are taken away.

3. Disposal:

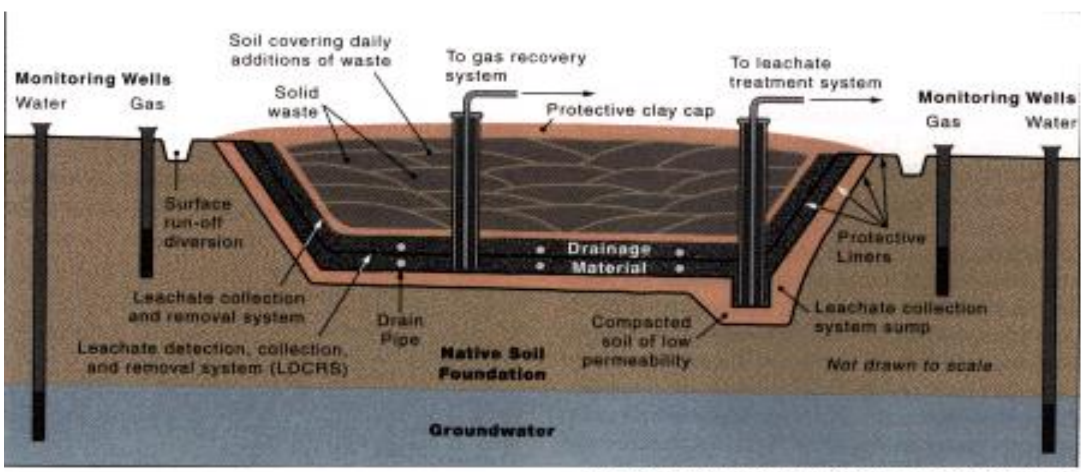
Once the various types of waste are collected, solid waste is disposed of according to type. Yard waste is often taken to a site where it's turned into mulch or compost, which may then be offered back to the public for use in gardens. Recyclables are bundled and either processed or sold to processing plants where they're turned back into usable materials. Hazardous wastes are taken to special dump sites reserved for their disposal. Regular trash that fits none of these categories but is still classified as solid waste is taken to a garbage dump or a land fill to be disposed of, depending on the community and its local rules and restrictions.

Industrial Solid waste:

Solid waste from small commercial and industrial areas is usually collected by handcart and taken directly to a temporary storage site. Larger commercial and industrial areas are serviced by trucks that transfer the waste directly to temporary or final disposal sites.

4.8.3. Methods for disposal of municipal and industrial solid waste:

i)Landfill: There are many different methods of disposing of waste. Landfill is the most common and probably accounts for more than 90% of the nation's municipal refuse even though Landfills have been proven contaminates of drinking water in certain areas. It is the most cost affective method of disposal, with collection and transportation accounting for 75% of the total cost. In a modern landfill, refuse is spread thin, compacted layers covered by a layer of clean earth. Pollution of surface water and groundwater is minimized by lining and contouring the fill, compacting and planting the uppermost cover layer, diverting drainage, and selecting proper soil in sites not subject to flooding or high groundwater levels. The best soil for a landfill is clay because clay is less permeable than other types of soil. Materials disposed of in a landfill can be further secured from leakage by solidifying them in materials such as cement, fly ash from power plants, asphalt, or organic polymers.

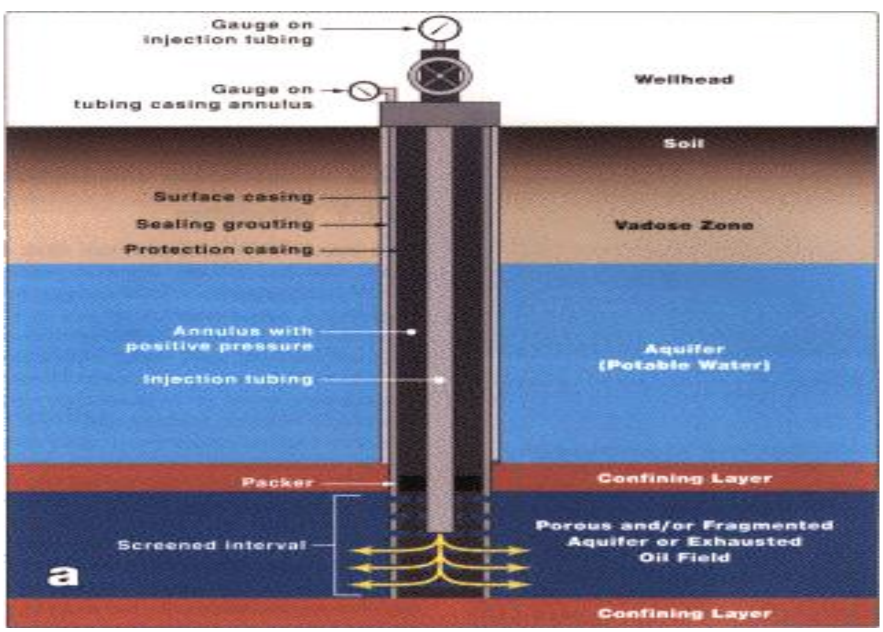


MODERN SANITARY LANDFILL

ii) Composting: Composting is a form of source reduction that involves the mixing of vegetable and organic refuse in order to speed the natural decomposition into fiber and micronutrients. For example, a compost pile in one's backyard recycles food scraps and lawn clippings that are deposited and mixed periodically. The decomposed product can then be used as fertilizer for the garden. Organic waste gives off energy in the form of heat when microorganisms metabolize the waste, causing it to lose between 40 and 75 percent of its original volume. After decontamination and refinement processes have been completed, the finished product is often used in landscaping, land reclamation, landfill cover, farming, and for nurseries. About one-quarter of household trash is organic material—food and yard waste.

iii)Incineration: Refuse is also burned in incinerators. It is more expensive but a safer method of disposal than landfills. Modern incinerators are designed to destroy at least 99.9% of the organic waste material they handle. Solid or liquid industrial wastes with a high organic content or containing chemicals such as polychlorinated biphenyls or other chlorinated hydrocarbons must be incinerated. Medical wastes are also incinerated in specially permitted facilities or commercial hazardous waste incinerators.
Incinerators are designed to destroy gases, liquids, solids, sludge, and slurries at high temperatures. One incineration process, the rotary kiln process, uses a cylindrical shell mounted on its side at a slight angle to the horizontal. As the kiln rotates, the waste moves down the slope, and the organic compounds burn. The kiln or burner is designed to withstand and maintain extremely high temperatures. Wastes are fed into the incinerator in batches or in a continuous stream.

Solid wastes may be fed into the incinerator in bulk or in containers using a conveyor or gravity system. Temperatures in the combustion chamber generally range from 1000 to 1400°C. There is a strong opposition to this method because of the apparent explosions and even earthquakes that have resulted from waste injection techniques.

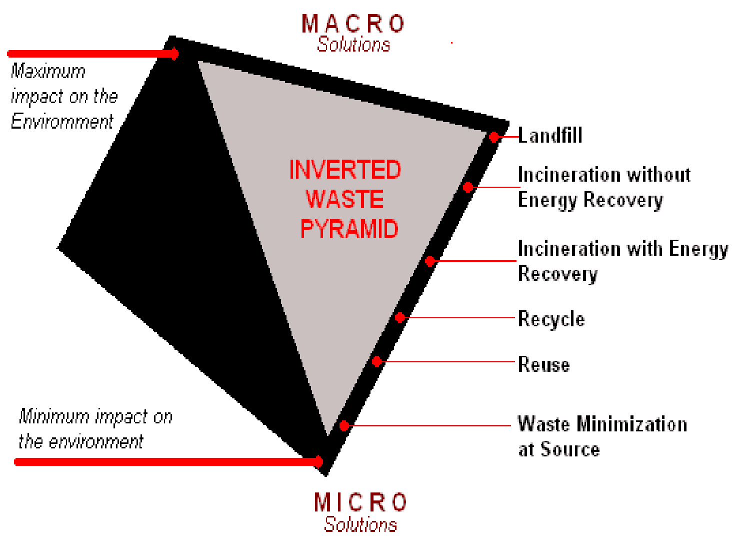


iv) Recovery: Recovery reduces the amount of garbage requiring disposal, thus saving landfill space and conserving the energy that would be used for incineration. Recovery also reduces environmental degradation and chemicals that pollute water resources, generates jobs and small-scale enterprise, reduces dependence on foreign imports of metals, and conserves water.

v) Recycling: Some analysts claim that more than half of consumer waste could be economically recycled. However, recycling sometimes requires more energy and water consumption than waste disposal. It depends on how far the materials must be transported and what is necessary to "clean" them before they can be reused. Demand for some recyclable materials is weak, making them economically unfeasible to recycle in a market-driven society.

4.8.4. Steps in solid waste Management:

- An integrated approach to the waste management is to be adopted. The waste management hierarchy includes following components. Reduce, Reuse, Recycle, Recover, Dispose.
- All above activities are arranged in a hierarchical manner. The first priority is waste avoidance, means not producing the waste. If the waste is produced then quantity should be minimized.
- The second priority is reuse i.e. maximizing recovery by reuse and recycling of suitable waste material. The three components i.e. reduce, reuse and recycle together is called waste prevention.
- Once the possibilities of waste prevention are exhausted, the next priority is reduced to the volume of residual wastes being passed on for final disposal i.e. extracting resources I the form of products or energy in the process.



4.8.5.Composition and characteristics of e- waste:

Common e-waste includes: PCs, TVs, telephones, cell phones, air conditioners, electronic toys, etc. E-waste composition changes from country to country.

The brominated(organ bromine compounds) flame retardants (BFRs) are used in computers, printed circuit boards, and in components such as connectors, in plastic covers, and in cables. BFRs are also used in a multitude of products, including, but not exclusively, plastic covers of television sets, carpets, paints, upholstery, and domestic kitchen appliances.

{The Cathode Ray Tube (CRT) is a vacuum tube containing an electron gun (a source of electrons) and a fluorescent screen, with internal or external means to accelerate and deflect the electron beam, used to create images in the form of light emitted from the fluorescent screen. The image may represent electrical waveforms (oscilloscope), pictures (television, computer monitor), radar targets and others.}

Hazardous components of electronic waste and where they are found:

Electronic waste or e-waste often has hazardous or toxic components that can have an impact on the environment once the materials end up in a landfill or if they are improperly managed and disposed. Below is a list of hazardous or toxic components of e-waste and where they may be found:

- Antimony trioxide - a flame retardant, added to cathode ray tube monitor (CRT) glass, found in printed circuit boards and cables
- Arsenic - in older cathode ray tubes and in light emitting diodes
- Barium - in the CRT
- Beryllium - often allied with copper to improve copper's strength, conductivity and elasticity. Old motherboards, contact springs found in printed circuit boards, relays, and in the mirror mechanism of laser printers. In power supply boxes which contain silicon controlled rectifiers and x-ray lenses
- Cadmium - circuit boards and semiconductors Rechargeable NiCd-batteries, fluorescent layer (CRT screens), printer inks and toners, photocopying-machines (printer drums)
- Chlorofluorocarbon (CFC) - Cooling unit, Insulation foam
- Chromium - in steel as corrosion protection, Data tapes, floppy-disks, circuit boards, photocopying-machines (printer drums)
- Cobalt - component in steel for structural strength and magnetivity
- Lead - cathode ray tubes, solder, batteries, printed wiring boards (circuit boards), solder on components
- Lithium - batteries
- Mercury - switches (mercury wetted) and housing, fluorescent lamps providing backlighting in liquid crystal displays (LCDs) for monitors and laptops, batteries, printed circuit boards
- Nickel - batteries, electron gun in CRT , printed circuit boards

- Polybrominated flame retardants (including polychlorinated biphenyls (PCB), polybrominated biphenyls (PBB), Polybrominated diphenyl ethers (PBDE), and tetrabromo bis-biphenol-a (TBBA))- plastic casings, cables, and circuit boards, condensers, transformers
- Polyvinyl chloride (PVC) - Cable insulation
- Selenium - circuit boards as power to supply rectifier, photocopying-machines (printer drums)
- Zinc - interior of CRT screens, printed circuit boards

4.8.6. Management of e- waste:

It is estimated that 75% of electronic items are stored due to uncertainty of how to manage it. These electronic junks lie unattended in houses, offices, warehouses etc. and normally mixed with household wastes, which are finally disposed off at landfills. This necessitates implement able management measures.

In industries management of e-waste should begin at the point of generation. This can be done by waste minimization techniques and by sustainable product design. Waste minimization in industries involves adopting:

- inventory management,
- production-process modification,
- volume reduction,
- recovery and reuse.

i) Inventory management: Proper control over the materials used in the manufacturing process is an important way to reduce waste generation. This can be done in two ways i.e. establishing material-purchase review and control procedures and inventory tracking system.

Developing review procedures for all material purchased is the first step in establishing an inventory management program. Procedures should require that all materials be approved prior to purchase. In the approval process all production materials are evaluated to examine if they contain hazardous constituents and whether alternative non-hazardous materials are available.

Another inventory management procedure for waste reduction is to ensure that only the needed quantity of a material is ordered. This will require the establishment of a strict inventory tracking system. Purchase procedures must be implemented which ensure that materials are ordered only on an as-needed basis and that only the amount needed for a specific period of time is ordered.

ii) Production-process modification: Changes can be made in the production process, which will reduce waste generation. This reduction can be accomplished by changing the materials used to make the product or by the more efficient use of input materials in the production process or both. Potential waste minimization techniques can be broken down into three categories:

- Improved operating and maintenance procedures,

- Material change and

- Process-equipment modification.

iii) Volume reduction: The techniques that can be used to reduce waste-stream volume can be divided into 2 general categories: source segregation and waste concentration. Segregation of wastes is in many cases a simple and economical technique for waste reduction. Wastes containing different types of metals can be treated separately so that the metal value in the sludge can be recovered. Concentration of a waste stream may increase the likelihood that the material can be recycled or reused. Methods include gravity and vacuum filtration, ultra filtration, reverse osmosis, freeze vaporization etc.

iv) Recovery and reuse: This technique could eliminate waste disposal costs, reduce raw material costs and provide income from a salable waste. Waste can be recovered on-site, or at an off-site recovery facility, or through inter industry exchange. A number of physical and chemical techniques are available to reclaim a waste material such as reverse osmosis, electrolysis, condensation, electrolytic recovery, filtration, centrifugation etc. For example, a printed-circuit board manufacturer can use electrolytic recovery to reclaim metals from copper and tin-lead plating bath.

4.8.7. Role of individual in prevention of pollution :

Pollution prevention is not a job ob of any specific person but all individuals must contribute in preventing pollution. Every individual should think about reducing pollution and act accordingly.

Important concepts to help reducing pollution:
1. Developing respect for all lives.
2. Importance of plantation of trees.
3. Reduce use of wood, paper etc. that comes from forests.
4. Reuse various useful materials.
5. Avoid using strong pesticides.
6. Prefer organic farming, avoid synthetic fertilizers.
7. Reduce consuming fossil fuels (oil, petrol, diesel, gas)
8. Save electricity use solar energy, rechargeable battery.
9. Avoid use of aerosol spray products or artificial air fresheners as they damages ozone layer.
10. Avoid use of non-degradable items e.g. plastic bags.
11. Separate garbage which can be recycled or reused.
12. Learn and educate local ecosystem and biodiversity.
13. Conserve water. Implement rain water harvesting.

14. Stop deforestation.
15. Control growth of population.

4.8.8. Pollution case studies:

1. Darkening effect of Taj Mahal:

Taj Mahal is one of the wonders of the world and is built of very superior kind of white marble. The Taj Mahal white marble is becoming yellowish or darkened slowly. The reason for decaying of Taj Mahal shine is identified as follows.

There is Mathura Oil refinery situated nearby Taj Mahal. The Mathura oil refinery emits sulphur dioxide (SO_2) in large quantity into the atmosphere. When rain water and oxygen (O_2) reacts with SO_2, it produces sulphuric acid (H_2SO_4) and sulphurous acid (H_2SO_3). This is called as acid rain.

$$H_2O \rightarrow H_2SO_3$$
$$SO_2 + O_2 \rightarrow SO_3$$
$$H_2O + SO_3 \rightarrow H_2SO_3$$

The sulphuric acid (acid rain) has corroding effect when it comes in contact with marble (CaCO3). This corroding effect results in darkening of marble (stone leprosy).

$$CaCO3 + H2SO4 \rightarrow CaSO4 + H2O$$

The Supreme Court has accepted the reason and ordered various polluting industries to either shut down or shift to other places.

2. Bhopal gas Tragedy:

An accident occurred on Dec'3, 1984, midnight, in a pesticide manufacturing company 'Union Carbide of India Ltd. UCIL) at Bhopal, M.P, India.

The UCIL manufactures pesticides by using methyle isocynate (MIC) gas, which is poisonous gas. The MIC gas is usually stored at a pressure of 8psi in storage tanker. That day pressure of tank no.610 abnormally increases from 8 psi to 40 psi. Because of this very high pressure safety valve (pressure release valve) gets failed and MIC gas was released in atmosphere with this pressure for at least two hours. More than 40 tons of MIC was leaked in liquid vapor form, spreading over 40 square kilometers of area.

Effects:

1. Because of deadly poisonous gas about 5000 peoples were killed.
2. Over 65000 people suffered from disorders in eyes, lungs, stomach, intestine, heart, kidney etc.
3. The indication of toxic effect bronchospam (coughing, chest pain and abdominal pain) was seen. Such symptoms were found in 25000 people and 2000 people died without treatment.
4. Nearly 1000 people reported blind.
5. About 1600 domestic animals were killed.
6. Many people could not walk because of nervous disorder.

4.9. BIO REMEDIATION:

Bioremediation consists of using living organisms (bacteria, fungi, actinomycetes, cyanobacteria and to a lesser extent, plants) to reduce or eliminate toxic pollutants. These organisms may be naturally occurring or laboratory cultivated. The physical, chemical and biological conditions of the culture environment have an influence on the shrimp to toxins like hydrogen sulphide, ammonia and carbon dioxide leads to stress and ultimately disease. Wastes produced in aquaculture farms differ in quality and quantity of components depending on the species farmed and the farming practices adopted. The wastes in aquaculture farms can be categorized as residual food and fecal matter, metabolic by-product, residues of biocides and bio-stats, fertilizer derived wastes and wastes produced during moulting and collapsing algal blooms.

The current approach to improving water quality in aquaculture is the application of microbes/enzymes to the ponds known as 'bioremediation'. When macro and microorganisms and/or their products are used as additives to improve water quality, they are referred to as bioremediators or bioremediating agents.

The newest attempt being made to improve water quality in aquaculture is the application of probiotics and enzymes to the ponds is known as bioremediation, which involves manipulation of microorganisms in ponds to enhance mineralization of organic matter and get rid of undesirable waste compounds.

Bioremediation may occur on its own (natural attenuation or intrinsic bioremediation) or may only effectively occur through the addition of fertilizers, oxygen, etc., that help encourage the growth of the pollution-eating microbes within the medium (bio stimulation). Microorganisms used to perform the function of bioremediation are known as bioremediators.

4.10. GLOBAL ENVIRONMENTAL PROBLEMS AND GLOBAL EFFORTS

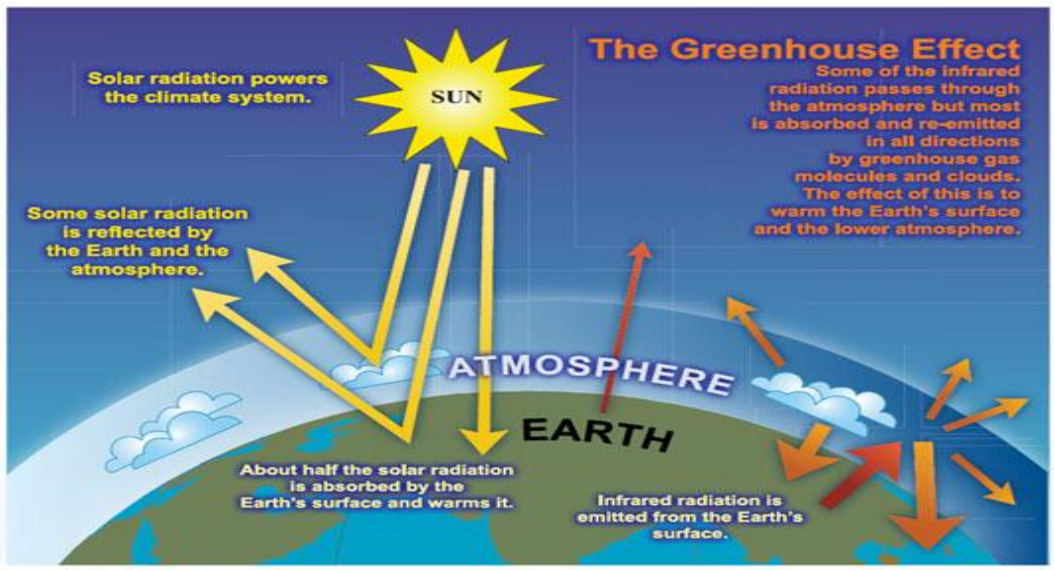
4.10.1. Green house effect:

The **greenhouse effect** is a process by which radioactive energy leaving a planetary surface is absorbed by some atmospheric gases, called greenhouse gases. They transfer this energy to other components of the atmosphere, and it is re-radiated in all directions, including back down towards the surface. This transfers energy to the surface and lower atmosphere, so the temperature there is higher than it would be if direct heating by solar radiation were the only warming mechanism.

This mechanism is fundamentally different from that of an actual greenhouse, which works by isolating warm air inside the structure so that heat is not lost by convection.

The green house effect may be defined as, “the progressive warming up of the earth’s surface due to blanketing effect of manmade CO_2 in the atmosphere”.

The Earth receives energy from the Sun in the form of visible light. This light is absorbed at the Earth's surface, and re-radiated as thermal radiation. Some of this thermal radiation is absorbed by the atmosphere, and re-radiated both upwards and downwards; that radiated downwards is absorbed by the Earth's surface. Thus the presence of the atmosphere results in the surface receiving more radiation than it would were the atmosphere absent; and it is thus warmer than it would otherwise be.



4.10.2. Green house gases:

Greenhouse gases are gases in an atmosphere that absorb and emit radiation within the thermal infrared range. This process is the fundamental cause of the greenhouse effect. The primary greenhouse gases in the Earth's atmosphere are water vapor, carbon dioxide, methane, nitrous oxide, and ozone. In our solar system, the atmospheres of Venus, Mars and Titan also contain gases that cause greenhouse effects. Greenhouse gases greatly affect the temperature of the Earth; without them, Earth's surface would be on average about 33 °C (59 °F) colder than at present.

The contribution to the greenhouse effect by a gas is affected by both the characteristics of the gas and its abundance. For example, on a molecule-for-molecule basis methane is about eighty times stronger greenhouse gas than carbon dioxide, but it is present in much smaller concentrations so that its total contribution is smaller. When these gases are ranked by their contribution to the greenhouse effect, the most important are:

Gas	Formula	Contribution(%)
Water Vapor	H_2O	36 – 72 %
Carbon Dioxide	CO_2	9 – 26 %
Methane	CH_4	4 – 9 %
Ozone	O_3	3 – 7 %

It is not possible to state that a certain gas causes an exact percentage of the greenhouse effect. This is because some of the gases absorb and emit radiation at the same frequencies as others, so that the total greenhouse effect is not simply the sum of the influence of each gas. The higher ends of the ranges quoted are for each gas alone; the lower ends account for overlaps with the other gases.

In addition to the main greenhouse gases listed above, other greenhouse gases include sulfur hexafluoride, hydro fluorocarbons and per fluorocarbons . Some greenhouse gases are not often listed. For example, nitrogen trifluoride has a high global warming potential (GWP) but is only present in very small quantities.

4.10.3. Global Warming:

What is global warming?

Global warming is when the earth heats up (the temperature rises). It happens when greenhouse gases (carbon dioxide, water vapor, nitrous oxide, and methane) trap heat and light from the sun in the earth's atmosphere, which increases the temperature. This hurts many people, animals, and plants. Many cannot take the change, so they die.

What is global warming doing to the environment?

Global warming is affecting many parts of the world. Global warming makes the sea rise, and when the sea rises, the water covers many low land islands. This is a big problem for many of the plants, animals, and people on islands. The water covers the plants and causes some of them to die. When they die, the animals lose a source of food, along with their habitat. Although animals have a better ability to adapt to what happens than plants do, they may die also. When the plants and animals die, people lose two sources of food, plant food and animal food. They may also lose their homes. As a result, they would also have to leave the area or die. This would be called a break in the food chain, or a chain reaction, one thing happening that leads to another and so on.

The oceans are affected by global warming in other ways, as well. Many things that are happening to the ocean are linked to global warming. One thing that is happening is warm water, caused from global warming, is harming and killing algae in the ocean.

Algae is a producer that you can see floating on the top of the water. (A producer is something that makes food for other animals through photosynthesis, like grass.) This floating green algae is food to many consumers in the ocean. (A consumer is something that eats the producers.) One kind of a consumer is small fish. There are many others like crabs, some whales, and many other animals. Fewer algae is a problem because there is less food for us and many animals in the sea.

Global warming is doing many things to people as well as animals and plants. It is killing algae, but it is also destroying many huge forests. The pollution that causes global warming is linked to acid rain. Acid rain gradually destroys almost everything it touches. Global warming is also causing many more fires that wipe out whole forests. This happens because global warming can make the earth very hot. In forests, some plants and trees leaves can be so dry that they catch on fire.

4.10.4. Effects of Global Warming

i) Sea level rise:

Sea levels in general are rising globally by about 3 mm (0.1181 inch) a year. Scientists blame rising temperatures caused by the growing amounts of greenhouse gases, such as carbon dioxide from burning fossil fuels, which trap heat in the atmosphere.

Oceans are absorbing a large part of this extra heat, causing them to expand and sea levels to rise. Warmer temperatures are also causing glaciers and parts of the ice blanketing Greenland and West Antarctica to melt.

Researchers found that sea-level rise is particularly high along the coastlines of the Bay of Bengal, the Arabian Sea, Sri Lanka, Sumatra and Java and that these areas could suffer rises greater than the global average.

But they also found that sea levels are falling in other areas. The study indicated that the Seychelles Islands and Zanzibar off Tanzania's coast show the largest sea-level drop.

Sea level rise is expected to increase flooding from the sea in large deltas and low-lying areas, more particularly in Africa, Asia (Bangladesh) and Europe (the Netherlands). Coastal inundation, typhoons, storm-surges, floods, erosion are also expected to increase.

Millions of people living in small islands and in or near coastal areas will be severely affected through loss of habitat and properties. Displacement of populations living in low lying areas represents a big challenge. This concerns especially developing countries which will have to bear a huge financial burden to relocate economic activities (agriculture, industry) and housings.

Another major consequence of sea level rise concerns water quality. Flooding by seawater will lead to salinization of groundwater that will have negative impacts on irrigation, aquaculture and fresh water availability. In some regions it will be necessary to invest in expensive desalinization techniques and to develop rainwater-harvesting systems.

ii) Climate change:

Climate change is a change in the statistical distribution of weather over periods of time that range from decades to millions of years. It can be a change in the average weather or a change in the distribution of weather events around an average (for example, greater or fewer extreme weather events). Climate change may be limited to a specific region, or may occur across the whole Earth.

In recent usage, especially in the context of environmental policy, climate change usually refers to changes in modern climate. It may be qualified as anthropogenic climate change, more generally known as "global warming" or "anthropogenic global warming.

Causes: Factors that can shape climate are climate forcing. These include such processes as variations in solar radiation, deviations in the Earth's orbit, mountain-building and continental drift, and changes in greenhouse gas concentrations. There are a variety of climate change feedbacks that can either amplify or diminish the initial forcing. Some parts of the climate system, such as the oceans and ice caps, respond slowly in reaction to climate forcing because of their large mass. Therefore, the climate system can take centuries or longer to fully respond to new external forcing.

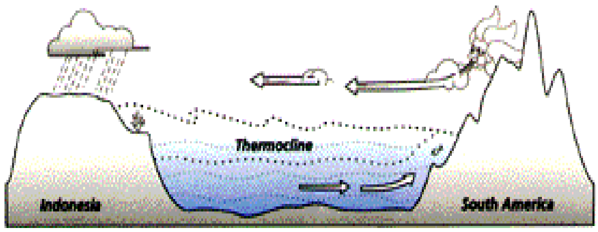
i) Plate tectonics: Over the course of millions of years, the motion of tectonic plates reconfigures global land and ocean areas and generates topography. This can affect both global and local patterns of climate and atmosphere-ocean circulation.

ii) Solar output: The sun is the predominant source for energy input to the Earth. Both long- and short-term variations in solar intensity are known to affect global climate.

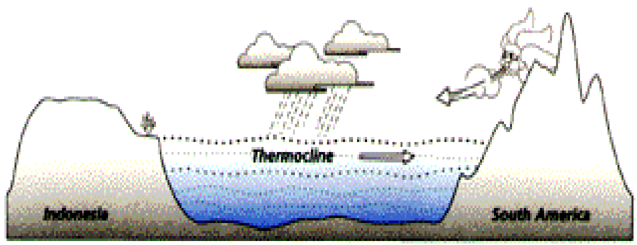
iii) Orbital variations: Slight variations in Earth's orbit lead to changes in the seasonal distribution of sunlight reaching the Earth's surface and how it is distributed across the globe. There is very little change to the area-averaged annually averaged sunshine; but there can be strong changes in the geographical and seasonal distribution.

iv) Gradual push, sudden shift: During a gradual push, things sometimes tip, slip, and break. Had the 1997 El Niňo(El Niño/La Niña-Southern Oscillation, or ENSO, is a quasi-periodic climate pattern that occurs across the tropical Pacific Ocean on average every five years, but over a period which varies from three to seven years. It is characterized by variations in the temperature of the surface of the tropical eastern Pacific Ocean - warming or cooling known as *El Niño* and *La Niña*) lasted twice as long, the rain forests of the Amazon basin and Southeast Asia could have quickly added much additional carbon dioxide to the air from burning and rotting, with heat waves and extreme weather quickly felt around the world.

To really understand the effects of an El Nino event, compare the normal conditions of the Pacific region and then see what happens during El Nino below.



Normal Conditions (Non El Nino)



El Nino Conditions

v) Volcanism: Volcanism is a process of conveying material from the crust and mantle of the Earth to its surface. Volcanic eruptions, geysers, and hot springs, are examples of volcanic processes which release gases and/or particulates into the atmosphere.

vi) Human influences: Anthropogenic factors are human activities that change the environment.

<u>Impacts of climate change on human environment:</u>

Many elements of human society and the environment are sensitive to climate variability and change. Human health, agriculture, natural ecosystems, coastal areas, and heating and cooling requirements are examples of climate-sensitive systems.

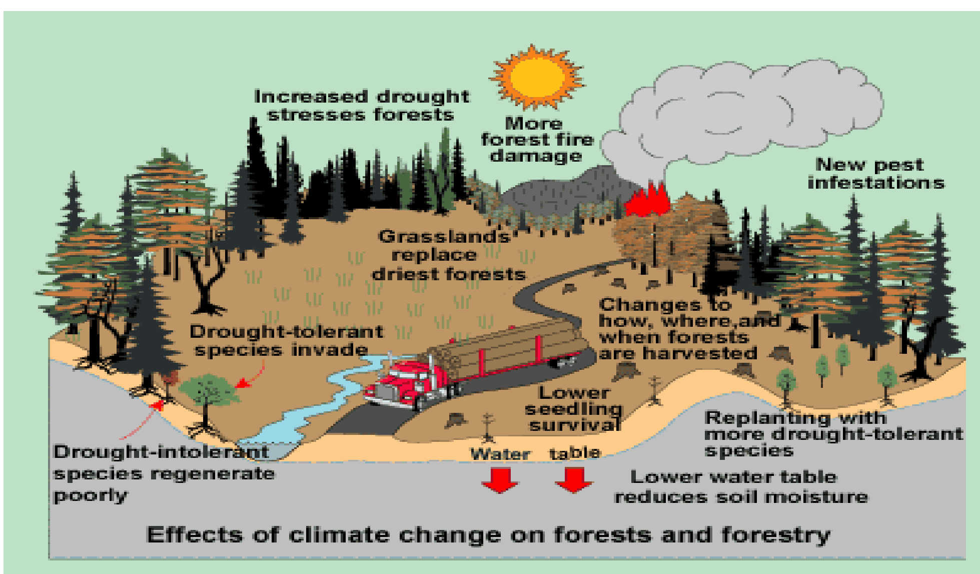
Rising average temperatures are already affecting the environment. Some observed changes include shrinking of glaciers, later freezing and earlier break-up of ice on rivers and lakes, lengthening of growing seasons, shifts in plant and animal ranges and earlier flowering of trees.

Health: Climate change may increase the risk of some infectious diseases, particularly those diseases that appear in warm areas and are spread by mosquitoes and other insects. These "vector-borne" diseases include malaria, dengue fever, yellow fever, and encephalitis. Also, algal blooms could occur more frequently as temperatures warm — particularly in areas with polluted waters — in which case diseases (such as cholera) that tend to accompany algal blooms could become more frequent.

Agriculture and food production: Changes in rainfall can affect soil erosion rates and soil moisture, both of which are important for crop yields. Agriculture is highly sensitive to climate variability and weather extremes, such as droughts, floods and severe storms.

Ecosystems and biodiversity: These changes can cause adverse or beneficial effects on species. For example, climate change could benefit certain plant or insect species by increasing their ranges. The resulting impacts on ecosystems and humans, however, could be positive or negative depending on whether these species were invasive (e.g., weeds or mosquitoes) or if they were valuable to humans (e.g., food crops or pollinating insects). The risk of extinction could increase for many species, especially those that are already endangered or at risk due to isolation by geography or human development, low population numbers, or a narrow temperature tolerance range.

Water quality: All regions of the world show an overall net negative impact of climate change on water resources and freshwater ecosystems. Areas in which runoff is projected to decline are likely to face a reduction in the value of the services provided by water resources. The beneficial impacts of increased annual runoff in other areas are likely to be tempered in some areas by negative effects of increased precipitation variability and seasonal runoff shifts on water supply, water quality and flood risks.



4.10.5. Measures to check global warming:

1. CO2 emissions can be cut by reducing the use of fossil fuels.
2. Implement energy conservation measures.
3. Utilize renewable resources such as wind, solar and hydropower.
4. Plant more trees.
5. Shift from coal to natural gas.
6. Adopt sustainable agriculture.
7. Stabilize population growth.
8. Efficiently remove CO2 from smoke staks.
9. Remove atmospheric CO2 by utilizing photosynthetic algae.

4.11. Ozone depletion:

The ozone layer protects the Earth from the ultraviolet rays sent down by the sun. If the ozone layer is depleted by human action, the effects on the planet could be catastrophic.

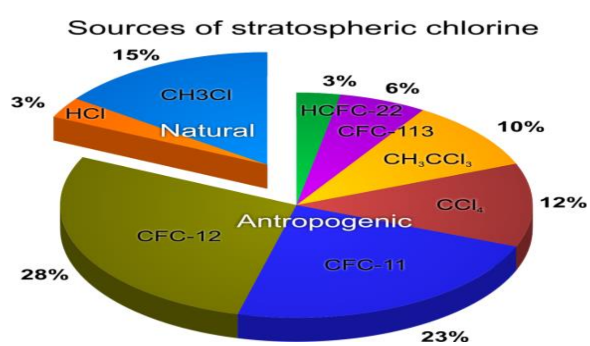
Ozone is present in the stratosphere. The stratosphere reaches 30 miles above the Earth, and at the very top it contains ozone. The sun's rays are absorbed by the ozone in the stratosphere and thus do not reach the Earth.

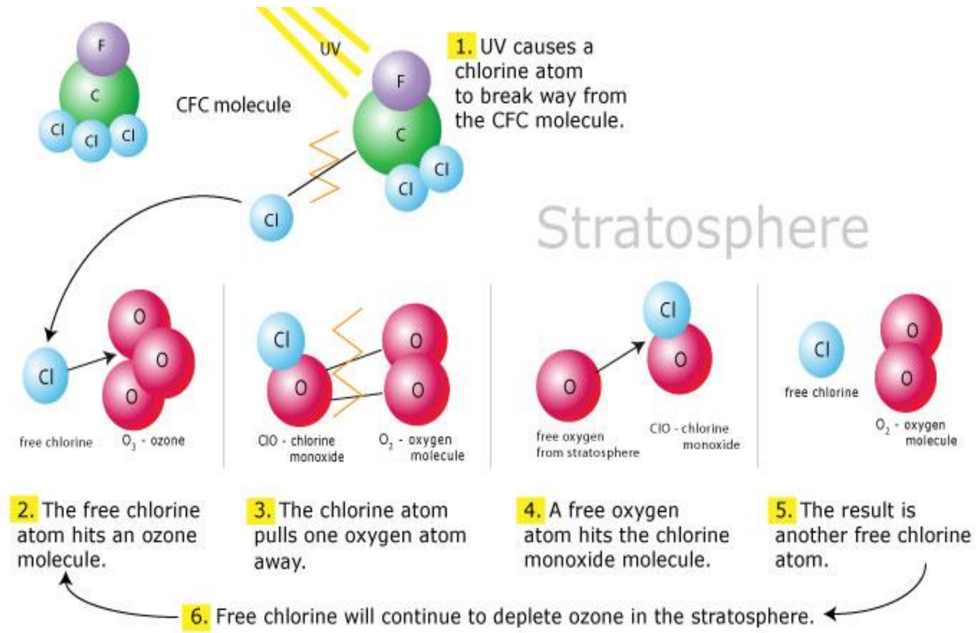
Ozone is a bluish gas that is formed by three atoms of oxygen. The form of oxygen that humans breathe in consists of two oxygen atoms, O_2. When found on the surface of the planet, ozone is considered a dangerous pollutant and is one substance responsible for producing the greenhouse effect. The highest regions of the stratosphere contain about 90% of all ozone.

The fact that the ozone layer was being depleted was discovered in the mid-1980s. The main cause of this is the release of CFCs, chlorofluorocarbons.

Antarctica was an early victim of ozone destruction. A massive hole in the ozone layer right above Antarctica now threatens not only that continent, but many others that could be the victims of Antarctica's melting icecaps. In the future, the ozone problem will have to be solved so that the protective layer can be conserved.

Causes: Only a few factors combine to create the problem of ozone layer depletion. The production and emission of CFCs, chlorofluorocarbons, is by far the leading cause.





Effects: Even minor problems of ozone depletion can have major effects. Every time even a small amount of the ozone layer is lost, more ultraviolet light from the sun can reach the Earth.

1. Every time 1% of the ozone layer is depleted, 2% more UV-B is able to reach the surface of the planet. UV-B increase is one of the most harmful consequences of ozone depletion because it can cause skin cancer.

2. For the fair skinned people lifelong exposure to the high level radiation of UV rays increases the risk of non-melamine skin cancer.

The increased cancer levels caused by exposure to this ultraviolet light could be enormous. The EPA estimates that 60 million Americans born by the year 2075 will get skin cancer because of ozone depletion. About one million of these people will die.

3. In addition to cancer, some research shows that a decreased ozone layer will increase rates of malaria and other infectious diseases. According to the EPA, 17 million more cases of cataracts can also be expected.

4. The environment will also be negatively affected by ozone depletion. The life cycles of plants will change, disrupting the food chain. Effects on animals will also be severe, and are very difficult to foresee.

5. Oceans will be hit hard as well. The most basic microscopic organisms such as plankton may not be able to survive. If that happened, it would mean that all of the other

animals that are above plankton in the food chain would also die out. Other ecosystems such as forests and deserts will also be harmed.

6. The planet's climate could also be affected by depletion of the ozone layer. Wind patterns could change, resulting in climatic changes throughout the world.

7. Degradation of paints, plastics and other polymeric material will result in economic loss due to effects of UV radiation.

8. The ozone depleting chemicals can contribute to the global warming i.e. increasing the average temperature of the earth's surface.

4.11.1. Ozone depleting substances:

Ozone depleting substances (ODS) are those substances which deplete the ozone layer and are widely used in refrigerators, air-conditioners, fire extinguishers, in dry cleaning, as solvents for cleaning, electronic equipment and as agricultural fumigants.

Ozone depleting substances include:

- Chlorofluorocarbons (CFCs)
- Halon (Halons are fully halogenated chemicals that have relatively long lifetimes in the atmosphere. They are broken down in the stratosphere releasing reactive bromine that is extremely damaging to ozone).
- Carbon tetrachloride, Methyl chloroform
- Hydrobromofluorocarbons (HBFCs)
- Hydrochlorofluorocarbons (HCFCs)
- Methyl bromide
- Bromochloromethane (BCM)

4.11.2. Control measures:

1. Replacing CFC's by other materials which are less damaging.
2. Use of gases such as methyl bromide which is a crop fumigant should be controlled.
3. Manufacturing and using of ozone depleting chemicals should be stopped.

4.12. Deforestation and desertification:

Deforestation is the clearance of forests by logging and/or burning (popularly known as slash and burn). Deforestation occurs for many reasons: trees or derived charcoal are used as, or sold, for fuel or as a commodity, while cleared land is used as pasture for livestock, plantations of commodities, and settlements. The removal of trees without sufficient reforestation has resulted in damage to habitat, biodiversity loss and aridity. It has adverse impacts on bio sequestration of atmospheric carbon dioxide. Deforested regions typically incur significant adverse soil erosion and frequently degrade into wasteland.

Disregard or ignorance of intrinsic value, lack of ascribed value, lax forest management and deficient environmental laws are some of the factors that allow deforestation to occur on a large scale. In many countries, deforestation is an ongoing issue that is causing extinction, changes to climatic conditions, desertification, and displacement of indigenous people.

Desertification is the degradation of land in arid and dry sub-humid areas due to various factors: including climatic variations and human activities. Desertification results chiefly from man-made activities. It is principally caused by overgrazing, over drafting of groundwater and diversion of water from rivers for human consumption and industrial use, all of these processes are fundamentally driven by overpopulation.

A major impact of desertification is reduced biodiversity and diminished productive capacity, for example, by transition from land dominated by shrub lands to non-native grasslands. For example, in the semi-arid regions of southern California, many coastal sage scrub and chaparral ecosystems have been replaced by non-native, invasive grasses due to the shortening of fire return intervals. This can create a monoculture of annual grass that cannot support the wide range of animals once found in the original ecosystem. In Madagascar's central highland plateau, 10% of the entire country has desertified due to slash and burn agriculture by indigenous peoples.

4.13. International conventions/ Protocols:

A **treaty** is an agreement under international law entered into by actors in international law, namely sovereign states and international organizations. A treaty may also be known as: **(international) agreement**, **protocol**, **covenant**, **convention**, **exchange of letters**, etc. Regardless of the terminology, all of these international agreements under international law are equally treaties and the rules are the same.

Earth summit:

The **United Nations Conference on Environment and Development** (UNCED), also known as the **Rio Summit**, **Rio Conference**, **Earth Summit** was a major United Nations conference held in Rio de Janeiro from 3 June to 14 June 1992.

In 1992, more than 100 heads of state met in Rio de Janeiro, Brazil for the first international Earth Summit convened to address urgent problems of environmental protection and socio-economic development. The assembled leaders signed the

Convention on Climate Change and the Convention on Biological Diversity, endorsed the Rio Declaration and the Forest Principles, and adopted Agenda 21, a 300 page plan for achieving sustainable development in the 21st century.

The issues addressed included:

- systematic scrutiny of patterns of production — particularly the production of toxic components, such as lead in gasoline, or poisonous waste including radioactive chemicals
- alternative sources of energy to replace the use of fossil fuels which are linked to global climate change
- new reliance on public transportation systems in order to reduce vehicle emissions, congestion in cities and the health problems caused by polluted air and smog
- the growing scarcity of water

An important achievement was an agreement on the Climate Change Convention which in turn led to the Kyoto Protocol. Another agreement was to "not carry out any activities on the lands of indigenous peoples that would cause environmental degradation or that would be culturally inappropriate".

The Commission on Sustainable Development (CSD) was created to monitor and report on implementation of the Earth Summit agreements. It was agreed that a five year review of Earth Summit progress would be made in 1997 by the United Nations General Assembly meeting in special session. This special session of the UN General Assembly took stock of how well countries, international organizations and sectors of civil society have responded to the challenge of the Earth Summit.

Kyoto Protocol:

is a protocol to the United Nations Framework Convention on Climate Change (UNFCCC or FCCC), aimed at fighting global warming. The UNFCCC is an international environmental treaty with the goal of achieving "stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system."

The Protocol was initially adopted on 11 December 1997 in Kyoto, Japan and entered into force on 16 February 2005. As of November 2009, 187 states have signed and ratified the protocol.

Under the Protocol, 37 industrialized countries commit themselves to a reduction of four greenhouse gases (GHG) (carbon dioxide, methane, nitrous oxide, sulphur hexafluoride) and two groups of gases (hydroflurocarbons and perflurocarbons) produced by them, and all member countries give general commitments. Countries agreed to reduce their collective greenhouse gas emissions by 5.2% from the 1990 level. Emission limits do not include emissions by international aviation and shipping, but are in addition to the industrial gases, chlorofluorocarbons, or CFCs, which are dealt with under the 1987 Montreal Protocol on Substances that Deplete the Ozone layer.

Details of the agreement

According to a press release from the United Nations Environment Program:

"The Kyoto Protocol is an agreement under which industrialized countries will reduce their collective emissions of greenhouse gases by 5.2% compared to the year 1990 (but note that, compared to the emissions levels that would be expected by 2010 without the Protocol, this target represents a 29% cut). The goal is to lower overall emissions from six greenhouse gases - carbon dioxide, methane, nitrous oxide, sulfur hexafluoride, HFCs, and PFCs - calculated as an average over the five-year period of 2008-12. National targets range from 8% reductions for the European Union and some others to 7% for the US, 6% for Japan, 0% for Russia, and permitted increases of 8% for Australia and 10% for Iceland."

Montreal protocol:

The Montreal Protocol on Substances That Deplete the Ozone Layer (a protocol to the Vienna Convention for the Protection of the Ozone Layer) is an international treaty designed to protect the ozone layer by phasing out the production of a number of substances believed to be responsible for ozone depletion. The treaty was opened for signature on September 16, 1987, and entered into force on January 1, 1989, followed by a first meeting in Helsinki (capital of Finland), May 1989. Since then, it has undergone seven revisions, in 1990 (London), 1991 (Nairobi), 1992 (Copenhagen), 1993 (Bangkok), 1995 (Vienna), 1997 (Montreal), and 1999 (Beijing). It is believed that if the international agreement is adhered to, the ozone layer is expected to recover by 2050. Due to its widespread adoption and implementation it has been hailed as an example of exceptional international co-operation with Kofi Annan quoted as saying that "perhaps the single most successful international agreement to date has been the Montreal Protocol". It has been ratified by 196 states.

Review questions:

1. Write a short note on
   a) Classification of pollutants
   b) Sources, effects of air pollution
2. Write a detailed note on control of air pollution.
3. Explain in detail sources, effects of water pollution.
4. Explain in detail control of water pollution.
5. a) Ambient air quality standards.
   b) Drinking water quality standards.
6. What is the impact of modern agriculture on soil.
7. a) Sources and effects of marine pollution.
   b) Control measures for soil pollution

8. Sources, effects and control measures for noise pollution.
9. Explain the terms 'thermal comfort' and 'heat islands effect'.
10. Write a short note on
    a) Solid waste classification
    b) Collection methods for municipal solid waste
11. Write a detailed note on disposal of solid waste.
12. Write a short note on
    a) Components of e- waste
    b) Management of e- waste.
13. Explain about bio remediation.
14. Write a detailed note on Greenhouse gases and greenhouse effect.
15. Detailed description on global warming.
16. Write a note on climate change and impacts on human environment.
17. Define Ozone layer depletion. How ozone layer is depleted by Ozone depleting substances.
18. Write a short note on
    a) Effects of Ozone layer depletion b) El-nino Phenomenon.
19. Write a detailed note on deforestation and desertification.
20. Write a detailed note on Earth Summit, Kyoto protocol and Montreal protocol.

Objective questions:

1. The most important indoor air pollutant is [d]
   a) SO_2 b) CO_2 c) NO_2 d) Radon gas

2. Damage to leaf structure by air pollutants causes [d]
   a) Dead areas of leaf b) Chlorophyll reduction
   c) Dripping of leaf d) All of these

3. Air pollutants mixing up with rain can cause [a]
   a) High acidity b) Low acidity c) Neutral conditions d) none of these

4. Industrial wastes may contain toxic [d]
   a) Chemicals b) Phenols c) Acids d) All of these
5. Dissolved oxygen in water comes from [d]
   a) Photosynthesis of aquatic plants b) Atmosphere
   c) None of these d) Both of these

6. Itai itai disease in Japan was caused by consumption of rice contaminated with [c]
   a) Mercury b) Iron c) Cadmium d) Zinc
7. Thermal pollutants can be controlled by [d]
   a) Cooling ponds b) Spray ponds c) Cooling towers d) All of these

8. Oil in water affects fish by affecting [a]

a) Gills b) Scales c) Eyes d) None of these

9. Which of the following have more penetration power? [c]

a) Alpha particles b) Beta particles c) Gamma – rays d) None of these

10. Bhopal gas tragedy occurred due to leakage of [a]

a) MIC b) DDT c) SO_2 d) Dioxins.

11. Which of the following enhance the frequency of earthquakes? [d]

a) Big dams b) Underground nuclear testing

c) Deep well disposal of liquid wastes d) All of these

12. The spread of the 'Stratosphere' above the Earth's surface is______ [b]

a) Below 15 km b) 10 to 50 km_ c) above 50 km d) above 100 km

13. The pungent smelling and light bluish gas in Stratosphere is due to_____ [b]

a) Carbon molecules b) Ozone molecules c) Nitrogen gas d) Oxygen

14. The ozone layer in the stratosphere acts as an efficient filter for ____ [a]

a) Solar UV- B rays b) X-rays c) Gamma rays d) UV-A rays

15. Ozone layer is found in_________ [c]

a) Ionosphere b) Troposphere c) Stratosphere d) All

16. The compounds that can easily break ozone molecules are______ [c]

a) Chlorine b) Bromine c) Both a & b d) None

17. CFC stands for___________ [a]

a) Chloro Fluro Carbons b) Carbon Fluorine Carbon

c) Compact Fluro Carbons d) None of the above

18. The main source of CFCs in household sector is___________ [b]

a) Televisions b) Refrigerators c) Washing machines d) All

19. The raise in present temperature of earth compared to 100 years ago is____ [b]

a) 2 – 6 ° C b) 0.3 – 0.6 ° C c) 10 ° C d) 0 ° C

20. Global warming is due to release of [b]

a) SO_2 b) greenhouse Gases c) inert gases d) free chlorine

21. The main constituents of Greenhouse gases (GHG) are on earth surface are [a]

a) CO2, CH_4 b) SOx c) nitrogen d) water vapor

22. The predicted raise in mean sea level due to global warming by the year 2100 [d]

a) 1 m b) 1 cm c) 2 m d) 9 -88 cm

23. The agency to look after the climate changes and for action to cut GHG ____ [a]

a) UNFCCC b) WHO c) DOE d) GOI

24. What is COP? [a]

a) Conference of Parties b) Coefficient of Pollution

c) Coalition of Parties d) Convention of People

25. The number of major greenhouse gases covered for their reduction by the Kyoto protocol are [c]

a) 10 b) 2 c) 6 d) 1

26. Which country emits maximum CO_2? [d]

a) Australia b) Iceland c) Norway d) USA

27. The year in which India ratified the Kyoto protocol____ [b]

a) 1997 b) 2002 c)2000 d)2003

28. CDM stands for [b]

a) Carbon Depletion Mechanism b) Clean Development Mechanism
c) Clear Development Mechanism d) Carbon Depletion Machinery

29. The fund intended to invest in projects that will reduce greenhouse gas emissions is___ [a]

a) Prototype Carbon Fund b) GHG Reduction Fund

c) Pollution Control Fund d) None

30. The first project GOI approved under prototype carbon fund is_____ [a]

a) SWERF b) NICE c) Asia-Urbs d) ICT

31. The name of the world commission on Environment and development is ____ [a]
  a) Brundtland Commission b) Zakaria Commission
  c) Planning Commission d) None of the above

FILL IN THE BLANKS

1. The main pollutants emitted by Thermal Power Plants are fly ash and SO_2
2. CO forms the highest proportion in the vehicular exhaust.
3. SO_2 during coal burning is produced due to oxidation of S contained in coal.
4. CO has affinity for hemoglobin 210 times more than oxygen.
5. Air pollutants affects plants by entering through stomata (leaf pores).
6. Sound frequency is expressed in Hz
7. Noise levels considered as threshold of pain are 140db.
8. As per Environmental (Protection) (Second Amendment) Rules. 1999 the permissible noise levels for fire-crackers are 5 log 10(N). (N= no. of crackers joined together)
9. Minamata disease occurred due to consumption of fish contaminated with Hg
10. Blue baby syndrome is caused by the presence of Nitrate in drinking water.
11. Power plants utilize only 1/3 of the energy provided by fossil fuel for their operation.
12. Radioactive strontium affects bones by depositing in the bones instead of Calcium.
13. The point where first movement occurs during earthquake is called epicenter
14. Various forms of cyclones are hurricanes, typhoons and willy willies

TRUE OR FALSE

1. Benzo- α- Pyrene of cigarette smoke is considered to cause cancer. **True**
2. SO2 does not affect respiratory passage. **False**
3. Washing of coal cannot remove sulphur. **False**
4. Sound can propagate without any medium. **False**
5. Rocket engine causes 90 db of noise. **False**
6. On Diwali the use of fire- crackers is not permitted between 6.00 pm and 10.00 pm. **False**
7. Groundwater is not less prone to contamination due to soil mantle. **False**
8. Groundwater contamination is irreversible. **True**
9. Waste heat in water does not affect the survival of sensitive species. **False**
10. B.O.D. is always higher than C.O.D **False**
11. High temperature increases the dissolved oxygen content in water. **False**
12. Solid waste material degraded by micro-organisms are called non-biodegradable. **False**
13. Floods leave everything contaminated whom flood water touches. **False**
14. Landslides occur when coherent rock of soil masses move down slope. **True**